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Seismic multiple suppression based on a deep neural network method for marine data

Kunxi Wang[1], Tianyue Hu[1], Shangxu Wang[2], and Jianxin Wei[2]

ABSTRACT

Seismic multiples in marine seismic data can affect the identification of oil and gas reservoirs. The efficiency of traditional multiple suppression methods, such as the Radon transform, depends on the accuracy of the velocity model for primaries and multiples, and the assumption of random background noise. To attenuate multiples with background noise, a method of primary reconstruction using a deep neural network based on data augmentation training is proposed. The designed deep neural network (DNN) includes convolutional encoding and decoding processes. The convolutional encoding process uses convolutional layers and maximum pooling layers to learn the features of the primaries, multiples, and background noise in a seismic data set. The convolutional decoding process uses these features to reconstruct the primaries and suppress the multiples and background noise. Using the data augmentation training method in the training phase, the full-wavefield data and the predicted multiples are added to background noise and then rotated to constitute the augmented data sets. This allows the DNN to have a better multiple suppression effect and better robustness. A well-trained DNN can be used for primary reconstruction in the same work area directly or in other work areas with a transfer learning method. Three examples of synthetic data with two simple models and a Pluto model verify the effectiveness, efficiency, stability, and good generalization of the proposed method for primary reconstruction and multiple suppression. Another example from field data finds that the proposed method can efficiently suppress seismic multiples under complex conditions.

INTRODUCTION

Multiples refer to the events that reflect more than once at the underground interface or the sea surface, which can distort the amplitude, frequency, and phase of the reflected waves from the target layer (Berkhout, 1982; Verschuur et al., 1992; Berkhout and Verschuur, 1997; Verschuur and Berkhout, 1997; Jiang et al., 2020; Li, 2020). Because conventional seismic data processing procedures are mostly based on the primary wavefield, the existence of multiples seriously affects velocity analysis, seismic migration, and tomography, thereby misleading the interpretation of seismic data (Weglein et al., 1997, 2003). Therefore, multiples are generally regarded as noise that needs to be suppressed and eliminated (Herrmann et al., 2008). There are two main types of multiple suppression methods: filtering methods (e.g., Hu and White, 1998; Ma et al., 2020) and wave-theory-based methods (e.g., Wiggins, 1988; Berkhout and Verschuur, 2005; Verschuur and Berkhout, 2005; Zhang et al., 2020).

Filtering methods predominantly use the differences in time and space characteristics of the primaries and multiples to suppress the multiples through different transformation methods. Predictive deconvolution is the earliest method used for the suppression of multiples (Robinson, 1967; Porsani and Ursin, 2007). However, the disadvantage of the predictive deconvolution method is that it is only suitable for suppressing short-period multiples in shallow seas and causes greater damage to the primary reflections. Using the difference in the motion correction velocities between the primaries and multiples, the over- and undercorrected events are classified as primaries and multiples, respectively. These classifications are based on the common-midpoint gather that has completed dynamic correction. Primaries and multiples can be effectively separated in the Radon domain using a parabolic Radon transform (Hampson,

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[1]Peking University, School of Earth and Space Sciences, Beijing, China. E-mail: kxwang@stu.pku.edu.cn; tianyue@pku.edu.cn (corresponding author).
[2]China University of Petroleum, State Key Laboratory of Petroleum Resources and Prospecting, Beijing, China. E-mail: wangsx@cup.edu.cn; weijx@cup.edu.cn.
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1986) owing to moveout differences. By eliminating primary coefficients in the Radon and time-space domains, multiples can be attenuated. Sacchi and Ulrych (1995) propose a high-resolution Radon algorithm based on Bayes' rule. In this algorithm, the Radon transform is considered a sparse inversion process, and the result is iteratively modified. With the use of 3D seismic data in seismic exploration, the 3D Radon (Hugonnet et al., 2008; Wang et al., 2017) method also has achieved a good multiple suppression effect (Ma et al., 2020). Another commonly used filtering method is beamform filtering (Hu and White, 1998). The adaptive beamformer is designed from models of primary and multiple reflection signals that have parametrically specified moveout and amplitude variations with offsets to estimate the signal and noise characteristics as part of the extraction process. However, when the primaries and multiples have similar dynamic correction speeds or when the horizontal changes in underground formations are large, the separation effect of the methods will be weakened.

Wave-theory-based methods predict and suppress multiples using multiple generation mechanisms. This broad category of methods can be divided into model-driven and data-driven methods, depending on the requirement of prior assumptions. Model-driven methods simulate multiples according to the wave equations associated with wavefield dynamic characteristics. The main example of which is the wavefield extrapolation method (Wiggins, 1988). Data-driven methods that do not require complete prior stratigraphic information can be applied to complex subsurface structures. These include the inverse-scattering series method (Weglein et al., 1997), surface-related multiple elimination (SRME) method (Verschuur et al., 1992; Verschuur and Berkhout, 1997), and virtual events method (Ikelle, 2006; Liu et al., 2018a). The SRME method is commonly used in industry to suppress multiples in marine data, in which large numbers of surface multiples exist. This method is data-driven and consists of two steps: first, prediction and then, subtraction of multiples (Verschuur et al., 1992; Verschuur and Berkhout, 1997). In the prediction stage, multiples are identified by a multidimensional convolution of the full-wavefield data. In the subtraction stage, the amplitude and phase of the predicted multiples are corrected by the matching operator to obtain estimated values for the true multiples. These values are subsequently subtracted from the signal, leaving the remaining primaries. Herrmann et al. (2008) present a sparse curvelet-domain subtraction approach by iteratively shrinking the curvelet coefficients under the framework of SRME. This process corrects amplitude errors by element-wise curvelet-domain scaling of the predicted multiples. Assuming that the primaries and multiples do not correlate locally in the time-space domain, Zhang et al. (2020) extract the leaked multiples from the initially estimated primaries. They did do this using local primary-and-multiple orthogonalization rather than restoring the damaged primaries to reduce the residue from multiples. However, these wave-theory-based methods also have some disadvantages. The amount of calculation for the inverse-scattering series method is quite large, which affects the practical application effect. The SRME and virtual events methods require data regularization, but the field data often require further interpolation due to insufficient spatial sampling, which impacts the processing effect. In addition, these two methods rely heavily on adaptive multiple subtraction algorithms, such as the minimum energy principle method (Verschuur and Berkhout, 1997), the L1-norm minimum single-channel matching method (Guitton and Verschuur, 2004), and the pattern learning matching method (Jiang et al., 2020; Li, 2020), etc. The selection of subtraction algorithms also determines the suppression effect on the multiples.

Recently, deep learning has made breakthroughs in image processing, natural language processing, speech recognition, and other fields (Hinton et al., 2006; Xiong et al., 2021). Deep learning method, based on data-driven methods, does not require the manual extraction of data features and can add prior knowledge to the model construction. This method can combine low-level features to form more abstract higher level features to represent attribute categories or features to better discover the effective feature representation of data and has the ability to generalize in other data sets. Deep learning has achieved better results in some fields than traditional methods. Typical deep learning structures include autoencoders (Hinton and Zemel, 1994), recurrent neural networks (Hochreiter and Schmidhuber, 1997), convolutional neural networks (LeCun et al., 1998), deep belief networks (Hinton et al., 2006), ResNet (He et al., 2016), U-net (Ronneberger et al., 2015), and generative adversarial networks (GANs) (Goodfellow et al., 2014).

Deep learning also has become an active field in exploration geophysics research. Huang et al. (2017) make progress in fault detection using convolutional neural network models. Mosser et al. (2018) use deep generative neural networks to model seismic velocity inversions. Wu et al. (2019) propose using artificial synthetic seismic data to train the convolutional neural network model and apply it to field data to perform automatic fault predictions. Das et al. (2019) propose a wave impedance inversion method based on a convolutional neural network. Wang and Hu (2022) use a deep neural network (DNN) trained in the common-shot gather and an iterative separation method to separate blended data (Wang and Mao, 2021). The adoption of DNNs for seismic and image noise elimination has become common practice in recent years. Without the need for data modeling and manually selected parameters, a denoising convolutional neural network (DnCNN) (Zhang et al., 2017a) was used to achieve high-quality denoising results of image data. The DnCNN model can handle Gaussian denoising at an unknown noise level. Liu et al. (2018b) propose a U-net seismic random noise suppression method based on data augmentation. This method uses a small amount of data from the artificial synthesis data set Pluto to generate a labeled data set. The preceding test, which achieved an excellent denoising effect on the Sigsbee data set, was an effective exploration of the generalization of convolutional neural networks. Yu et al. (2019) test the feasibility of DNN in random noise and linear noise elimination and analyze several general problems regarding the understanding of complicated networks, empirical hyperparameter settings, and intense computation during training. For multiple eliminations, Siahkoohi et al. (2019a) initially use SRME and the estimation of primaries by sparse inversion methods to preprocess the original data and subsequently use the original data in combination with preprocessed primary data to train the generative adversarial network. The trained DNN eliminates multiples more effectively. Li and Gao (2020) use a convolutional neural network to extract multifeatures of the predicted multiples and use features to match the multiples, which enable the balancing of the removal of multiples and the retention of primaries.

In this study, we have used a deep neural network based on data augmentation training (DNNDAT) to attenuate multiples. The designed DNN includes convolutional encoding and decoding processes. Using the trained DNN parameters, the convolutional encoding extracts continuous signal features in the seismic data and distinguishes between primaries and multiples. The convolutional

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decoding uses the features of the primaries extracted during encoding to reconstruct the separated data. Considering that the training set of the DNN training phase is often small, which easily causes the DNN to overfit, we add background noise of different intensities to the input data of the training set and use rotation to obtain an augmented data set. The field data contain a large amount of background noise, and the augmented data set can further improve the robustness capability of the DNN. We use transfer learning to allow the DNN parameters of one work area to be accurately applied in another work area, which further improves the generalization performance of the DNNDAT method. The effectiveness, stability, and generalization of the proposed method in the primary reconstruction and multiple suppression are demonstrated by four examples: two simple models, the Pluto model, and one set of field data.

METHOD

Review of SRME

Surface multiple suppression methods for marine seismic data include SRME. The SRME method is divided into the prediction and adaptive subtraction steps described previously. Based on the primary wavefield measurements (Berkhout, 1982), the wavefield excited by a conventional seismic source is subjected to impulse response $\boldsymbol{\delta}$ of the underground medium, and part of the energy is transmitted to the surface and received by the detector $\mathbf{D}$ to obtain the primaries $\mathbf{G}_0$. The process of generating primaries from the source wavefield through reflection can be expressed as

$$\mathbf{G}_0 = \mathbf{D}\boldsymbol{\delta}\mathbf{S}^+, \tag{1}$$

where $\mathbf{S}^+$ is the source output.

The upgoing wavefield $\mathbf{G}^-$ can form a new downgoing wavefield $\mathbf{R}^-\mathbf{G}^-$ after the action of the reflection operator $\mathbf{R}^-$ at a free interface. If $\mathbf{R}^-\mathbf{G}^-$ is regarded as the secondary source, a free-surface multiple $\mathbf{U}_0$ can be obtained after rereflection:

$$\mathbf{U}_0 = \mathbf{D}\boldsymbol{\delta}\mathbf{R}^-\mathbf{G}^-. \tag{2}$$

The entire downgoing wavefield $\mathbf{G}^+$ can be expressed as

$$\mathbf{G}^+ = \mathbf{S}^+ + \mathbf{R}^-\mathbf{G}^-. \tag{3}$$

The full-wavefield $\mathbf{G}$ received by receivers can be expressed as

$$\mathbf{G} = \mathbf{G}_0 + \mathbf{U}_0 = \mathbf{D}\boldsymbol{\delta}\mathbf{S}^+ + \mathbf{D}\boldsymbol{\delta}\mathbf{R}^-\mathbf{G}^-. \tag{4}$$

Using the free interface operator $\mathbf{A} = (\mathbf{S}^+)^{-1}\mathbf{R}^-\mathbf{D}^{-1}$ and the full-wavefield operator $\mathbf{G} = \mathbf{D}\mathbf{G}^-$, equation 4 can be rewritten as

$$\mathbf{G} = \mathbf{G}_0 + \mathbf{G}_0\mathbf{A}\mathbf{G}. \tag{5}$$

Therefore, the surface multiples can be suppressed iteratively using the prediction-subtraction method:

$$\begin{cases} \mathbf{G}_0^{(n+1)} = \mathbf{G} - \mathbf{G}_0^{(n)}\mathbf{A}\mathbf{G} = \mathbf{G} - A^{(n+1)}\mathbf{U}^{(n+1)} \\ \mathbf{G}_0^{(0)} = \mathbf{G} \end{cases}, \tag{6}$$

where n is the number of iterations, $\mathbf{U}^{(n+1)} = \mathbf{G}_0^{(n)}\mathbf{G}$ represents the predicted multiples, and A is the matching operator.

DNN structure

The convolutional layer that can process 2D data directly has local perception and parameter sharing characteristics. Each neuron does not need to perceive the pixels in all of the data but only the local data. The neuron subsequently combines these local data into a higher layer to obtain the data characteristics. To make full use of the spatial continuity features of 2D seismic data and avoid problems in training and overfitting (Wu et al., 2019), the DNN framework in this study uses the convolutional layer to learn data features. Its structure is illustrated in Figure 1. The DNN is a style of supervised learning that is divided into two parts: a convolutional encoder and a convolutional decoder. In convolutional encoding, 14 convolutional and four downsampling layers with a pool size of two grasp the temporal and spatial positional relationship of seismic data and learn the sparse expression characteristics of the primaries and multiples in the training set. Subsequently, in convolutional decoding, 14 convolutional layers and four upsampling layers with a ratio of two map the low-dimensional features containing the primary or multiple signals back to the high-dimensional space enabling signal reconstruction. Simultaneously, the feature fusion (Concat) in the DNN combines the features obtained by the downsampling layers during convolutional encoding with those obtained during convolutional decoding. This can retain more high-level feature map information, obtain more accurate seismic temporal and spatial coherence features, and achieve better multiple suppression effects. To improve the generalization performance of the DNN, we use dropout (Srivastava et al., 2014) after the 12th and 15th convolutional layers to randomly discard 50% of the neural units. The size of all convolution kernels (filters) is 3×3, except for the 16th, 19th, 22nd, and 25th convolutional layers, whose convolution kernel size is 2×2. Moreover, the numbers of the convolution kernels shown in Figure 1 are 32, 32, 32, 64, 128, 128, 128, 256, 256, 256, 512, 512, 1024, 1024, 1024, 512, 512, 256, 256, 256, 128, 128, 128, 64, 64, 64, 32, and 3. To maintain the size of feature maps after each convolution operation, we perform the same padding processing to fill map peripheries with zero (Wang and Hu, 2022).

Data augmentation training method

In the DNN structure (Figure 1), the input data (left side) are marked as $\mathbf{X}^A$, and the ground truth (right side) is denoted as $\mathbf{Y}^A$ (also known as label data). The input data $\mathbf{X}^A$ are composed of two channels: the first is the full-wavefield data and the second is the predicted multiples through convolution (Verschuur et al.,

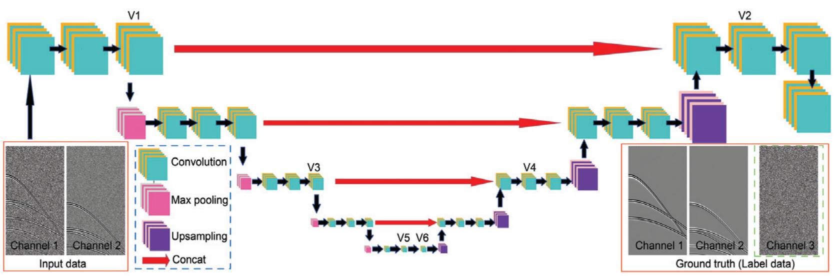

Figure 1. Structure of the DNN. The V1–V6 layers are used to obtain the visualization results.

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1992). The combination of the input data and ground truth forms the training data set pair $\{\mathbf{X}^A, \mathbf{Y}^A\}$. Before training the DNN, the amplitudes of the training sets are normalized to [−1,1].

In the general training (GT) method, the ground truth is composed of two channels: true primaries and true multiples. In collected data, the common-shot gather of the true full wavefield containing multiples is recorded as $\mathbf{F}$; and the corresponding common-shot gather of the predicted multiples is denoted as $\mathbf{C}$, the true primaries are recorded as a data set $\mathbf{P}$, and the common-shot gather of the true multiples is recorded as a data set $\mathbf{M}$. In the DNN training stage, $N1$ common-shot gathers are extracted from the data set $\mathbf{F}$, $\mathbf{C}$, $\mathbf{P}$, and $\mathbf{M}$ at equal intervals according to the same shot spacing, which are, respectively, denoted as $\mathbf{F}_k, \mathbf{C}_k, \mathbf{P}_k, \mathbf{M}_k$, $k = 1, 2, 3, \cdots, N1$. Here, the input data are $\langle \mathbf{F}_k; \mathbf{C}_k \rangle$, and the ground truth is $\langle \mathbf{P}_k; \mathbf{M}_k \rangle$. These are combined into the GT data set pairs $\{\langle \mathbf{F}_k; \mathbf{C}_k \rangle, \langle \mathbf{P}_k; \mathbf{M}_k \rangle\}$. Using GT data set pairs to train a DNN is called the GT method.

When using a deep neural network based on the general training (DNNGT) method to suppress multiples, small samples are often encountered. The main problem faced by learning with small samples is that the number of supervised samples also is too small and the DNN parameters require sufficient data to support training (Wong et al., 2016; Dao et al., 2019). However, in field data applications, it is difficult to obtain a large amount of label data. When the amount of data in the training set is small, overfitting tends to occur. Rotating a small sample learning data set to increase the amount of data is a relatively straightforward solution. In addition to the lack of label data, uneven data distribution commonly leads to poor model robustness and performance due to overfitting (Perez and Wang, 2017; Shorten and Khoshgoftaar, 2019). In addition, the field data often contain a certain amount of noise. Adding background noise with different amplitudes to the input data can increase the robustness of the DNN. Therefore, the data augmentation method of adding background noise and rotation to the training data set can improve the performance of primary and multiple separation of the DNN.

To prevent overfitting and further improve the generalization ability of the DNN, noise with different intensities is added to the input data $\mathbf{X}^A$ to obtain an augmented data set. When training the DNN, added background noise also is used as the third channel of the ground truth. Based on the GT data set pairs, we add $N2$ background noise $\mathbf{n}1_k^u$ and $\mathbf{n}2_k^u$ with different intensities to $\mathbf{F}_k$ and $\mathbf{C}_k$, respectively, and $u = 1, 2, 3, \cdots, N2$. When $u = 1$, no background noise is added to the input data, that is

$$\begin{cases} \mathbf{n}1_k^1 = 0 \\ \mathbf{n}2_k^1 = 0 \end{cases}. \tag{7}$$

The input data $\mathbf{X}^A$ are obtained by adding noise $\mathbf{n}1_k^u$ and $\mathbf{n}2_k^u$ with different intensities to the full-wavefield $\mathbf{F}_k$ and predicted multiples $\mathbf{C}_k$. The true primaries $\mathbf{P}_k$, true multiples $\mathbf{M}_k$, and background noise $\mathbf{n}1_k^u$ form the three channels of ground truth $\mathbf{Y}^A$, namely

$$\mathbf{Y}^A = \langle \mathbf{P}_k^u; \mathbf{M}_k^u; \mathbf{n}1_k^u \rangle. \tag{8}$$

Therefore, the training data set after adding noise is $\{\langle \mathbf{F}_k^u; \mathbf{C}_k^u \rangle, \langle \mathbf{P}_k^u; \mathbf{M}_k^u; \mathbf{n}1_k^u \rangle\}$, where

$$\begin{cases} \mathbf{F}_k^u = \mathbf{F}_k + \mathbf{n}1_k^u \\ \mathbf{C}_k^u = \mathbf{C}_k + \mathbf{n}2_k^u \\ \mathbf{P}_k^u = \mathbf{P}_k \\ \mathbf{M}_k^u = \mathbf{M}_k \end{cases}. \tag{9}$$

When the values of k are the same, the input data contain the same full wavefield and predicted multiples, for which the added background noise is different. The corresponding label data contain the same true primaries, true multiples, and different intensities of background noise.

To improve the ability of the predicted multiples to calibrate the true multiples and to avoid obtaining a single small training data, we rotate the training data set $\{\langle \mathbf{F}_k^u; \mathbf{C}_k^u \rangle, \langle \mathbf{P}_k^u; \mathbf{M}_k^u; \mathbf{n}1_k^u \rangle\}$ at a certain angle to obtain an augmented training data set pair $\{\langle \mathbf{F}_{k,h}^u; \mathbf{C}_{k,h}^u \rangle, \langle \mathbf{P}_{k,h}^u; \mathbf{M}_{k,h}^u; \mathbf{n}1_{k,h}^u \rangle\}$, where

$$\begin{cases} \mathbf{F}_{k,h}^u = \kappa_h \mathbf{F}_k^u \\ \mathbf{C}_{k,h}^u = \kappa_h \mathbf{C}_k^u \\ \mathbf{P}_{k,h}^u = \kappa_h \mathbf{P}_k^u \\ \mathbf{M}_{k,h}^u = \kappa_h \mathbf{M}_k^u \\ \mathbf{n}1_{k,h}^u = \kappa_h \mathbf{n}1_k^u \end{cases}. \tag{10}$$

Here, κ_h refers to rotation of the data set h times at a fixed angle interval starting from 0° and appending the data obtained by the rotation to the existing data. Using augmented training data set pairs to train a DNN is called the data augmentation training (DAT) method. Figure 2 shows a flowchart of making the GT data set pairs and the augmented training data set pairs for the DNN training.

DNN training

When training the DNN, each network layer contains convolution operations as follows (Jain et al., 1996):

$$\mathbf{Y}_j^L = \sum_i^I \mathbf{W}_j^L * \mathbf{X}_i^{L-1} + \mathbf{b}_j^L, \quad j = 1, 2, \cdots, J, \tag{11}$$

where $\mathbf{Y}_j^L$ represents the jth feature map extracted from the Lth layer, $\mathbf{X}_i^{L-1}$ represents the feature map corresponding to the ith convolution kernel of the $(L-1)$th convolutional layer, I represents the total number of feature maps of the $(L-1)$th layer, $\mathbf{W}_j^L$ represents the jth convolution kernel of the Lth convolution layer, also known as the weight matrix, J is the total number of convolution kernels in the Lth layer, and $\mathbf{b}_j^L$ represents the bias coefficient corresponding to the jth convolution kernel in the Lth layer.

Because the convolution operation is linear, nonlinear mapping is required through the activation function σ (Han et al., 1996):

$$\mathbf{Y}_j^L = \sigma\left(\sum_i^I \mathbf{W}_j^L * \mathbf{X}_i^{L-1} + \mathbf{b}_j^L\right), \quad j = 1, 2, \cdots, J. \tag{12}$$

In the first 27 convolutional layers, the activation function is the rectified linear unit (ReLU) function (Wu et al., 2019). Unilateral inhibition of the activation function can effectively maintain the sparsity of neurons and is defined as

$$\mathrm{ReLU}(x) = \max\{0, x\} = \begin{cases} x, & \text{if } x \geq 0 \\ 0, & \text{if } x < 0 \end{cases}. \tag{13}$$

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In other words, when x is less than zero, it is in an inactive state, and when x is greater than zero, it is in an active state. The ReLU function can avoid gradient disappearance and explosion and greatly simplify the calculation of backpropagation (Yu et al., 2019). Because the distribution range of the output data values is $[-1,1]$ in the final convolutional layer, tanh is the activation function (LeCun et al., 1998):

$$\tanh(x) = \frac{\sinh(x)}{\cosh(x)} = \frac{e^x - e^{-x}}{e^x + e^{-x}}. \tag{14}$$

Maximum pooling and upsampling do not need to learn DNN parameters, so we only need to determine the parameters $\boldsymbol{\theta} = \{\mathbf{W}_j^L, \mathbf{b}_j^L\}$ in convolution operations. If the DNN is denoted as $\mathcal{F}$, then $\mathcal{F}(x, \boldsymbol{\theta})$ represents the DNN output under parameters $\boldsymbol{\theta}$ when the input data are x. We define the mean absolute error (MAE) as a loss function to describe the matching degree between the training model and the training sample (Zhao et al., 2017). The loss functions of the DNNGT and DNNDAT methods can be expressed as follows:

$$\mathrm{MAE}(\boldsymbol{\theta}) = \frac{1}{N}\sum_{v=1}^{N} \|\mathcal{F}(\mathbf{X}_v^A, \boldsymbol{\theta}) - \mathbf{Y}_v^A\|^1, \tag{15}$$

where N represents the number of training sample pairs. A good optimization operator is beneficial for the iterative convergence of training (Guo et al., 2020; Luo et al., 2020; Xiao et al., 2020). The Adam optimization algorithm (Wang and Hu, 2022) is used to minimize the loss function. This algorithm designs independent adaptive learning rates for different parameters, which is undertaken by calculating the first- and second-order moment estimations of the gradient (Kingma and Ba, 2014). The Adam algorithm can achieve a high computational efficiency while maintaining low-memory requirements.

DNN testing

During DNN testing, noisy or clean full-wavefield data are used for the input data $\mathbf{X}^T$. Using the DNNGT method with parameter $\boldsymbol{\theta}^t$, the reconstructed results are

$$\langle \mathbf{P}^t; \mathbf{M}^t \rangle = \mathcal{F}(\mathbf{X}^T, \boldsymbol{\theta}^t). \tag{16}$$

The results $\mathbf{P}^t$ and $\mathbf{M}^t$ are the reconstructed primaries and multiples, respectively. Using the DNNDAT method with parameters $\boldsymbol{\theta}^s$, the reconstructed results are

$$\langle \mathbf{P}^s; \mathbf{M}^s; \mathbf{n}^s \rangle = \mathcal{F}(\mathbf{X}^T, \boldsymbol{\theta}^s), \tag{17}$$

where $\mathbf{P}^s$, $\mathbf{M}^s$, and $\mathbf{n}^s$ are the reconstructed primaries, multiples, and background noise, respectively.

After obtaining the result using DNNGT, the separated and reconstructed primaries can be obtained using three different modes:

$$\mathbf{P} - \mathrm{GT1} = \mathbf{P}^t, \tag{18}$$

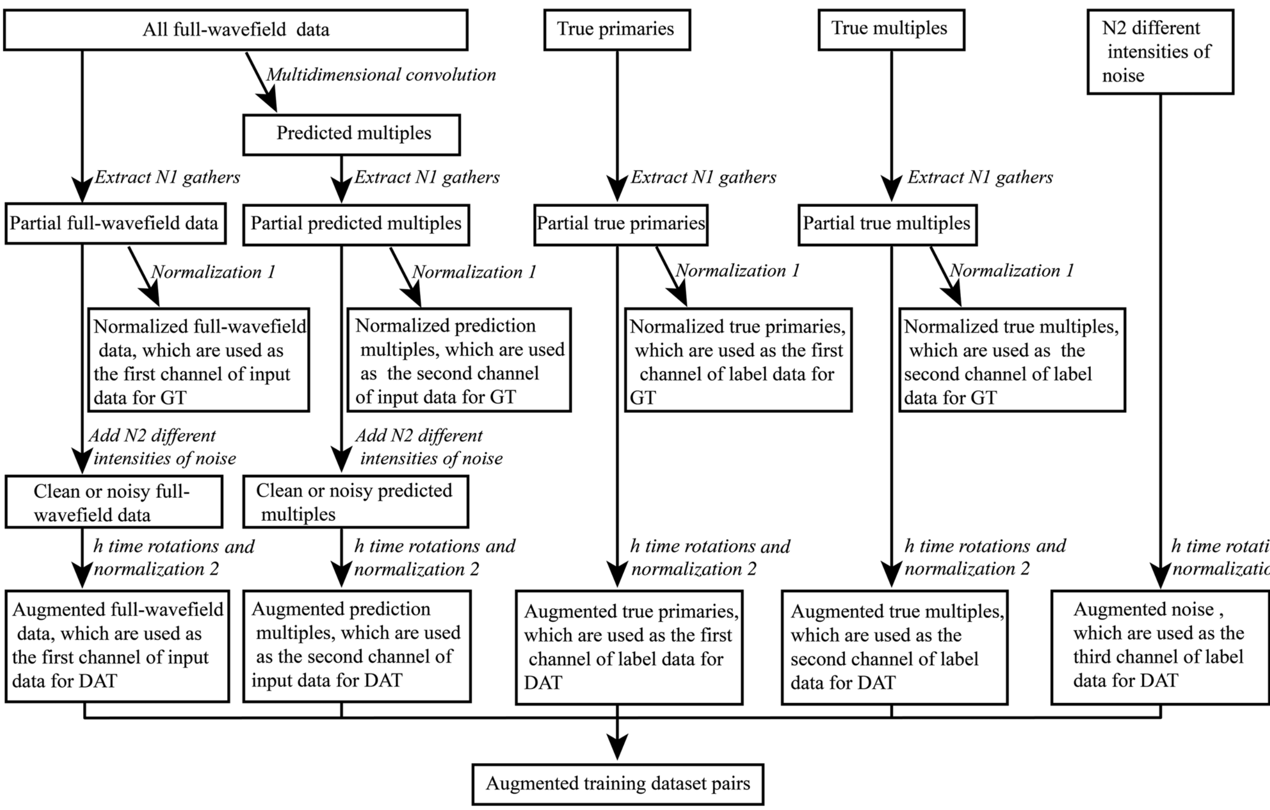

Figure 2. Flowchart showing the construction of the GT and augmented training data set pairs.

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$$\mathbf{P}-\mathrm{GT2}=\mathbf{X}^{T1}-\mathbf{M}^{t}, \tag{19}$$

$$\mathbf{P}-\mathrm{GT3}=\frac{(\mathbf{P}^{t}+(\mathbf{X}^{T1}-\mathbf{M}^{t}))}{2}, \tag{20}$$

where $\mathbf{X}^{T1}$ represents the first channel of the input data $\mathbf{X}^{T}$. Equations 18–20 are used to calculate the reconstructed primaries via the DNNGT method and are called the **P**-GT1, **P**-GT2, and **P**-GT3 modes, respectively. Because the DNNDAT can obtain the prediction results of three different channels, the primaries in the noisy or clean full-wavefield data can be estimated using three different modes:

$$\mathbf{P}-\mathrm{DAT1}=\mathbf{P}^{s}, \tag{21}$$

$$\mathbf{P}-\mathrm{DAT2}=\mathbf{X}^{T1}-\mathbf{M}^{s}-\mathbf{n}^{s}, \tag{22}$$

$$\mathbf{P}-\mathrm{DAT3}=\frac{(\mathbf{P}^{s}+(\mathbf{X}^{T1}-\mathbf{M}^{s}-\mathbf{n}^{s}))}{2}. \tag{23}$$

We call the modes of primary separation and reconstruction obtained using equations 21–23, **P**-DAT1, **P**-DAT2, and **P**-DAT3 modes, respectively.

EXAMPLE

Simple model one

A simple velocity model consisting of four horizontal layers is shown in Figure 3. Using finite-difference forward modeling, 500 common-shot gathers of the full-wavefield data (**F**), corresponding predicted multiples (**C**), true primaries (**P**), and true multiples (**M**) are obtained. Reflection responses are recorded on the surface using 500 receivers. The source and receiver spacings are 10 m, the sampling interval is 4 ms, and the central frequency of the seismic source is 15 Hz.

In the DNN training stage, 5% common-shot gathers are extracted from data sets **F**, **C**, **P**, and **M** at intermediate intervals, namely N1 = 25, to obtain the GT set pairs. To increase the amount of training data and improve the robustness performance of the DNN, eight different white Gaussian noise intensities are used as background noise. These are added to input data $\mathbf{F}_k$ and $\mathbf{C}_k$ in the GT data set pairs, namely, N2 = 8. Then, we rotate the noise-added GT data set pairs seven times at a 45° angle to obtain the augmented training data set pairs $\{\langle\mathbf{F}^{u}_{k,h};\mathbf{C}^{u}_{k,h}\rangle,\langle\mathbf{P}^{u}_{k,h};\mathbf{M}^{u}_{k,h};\mathbf{n}1^{u}_{k,h}\rangle\}$. The total numbers of training pairs used for the GT and DAT are 25 and 1600, respectively. Figure 4 shows part of the augmented training data set pairs: $\{\langle\mathbf{F}^{1}_{3,0};\mathbf{C}^{1}_{3,0}\rangle,\langle\mathbf{P}^{1}_{3,0};\mathbf{M}^{1}_{3,0};\mathbf{n}1^{1}_{3,0}\rangle\}$, $\{\langle\mathbf{F}^{1}_{3,1};\mathbf{C}^{1}_{3,1}\rangle,\langle\mathbf{P}^{1}_{3,1};\mathbf{M}^{1}_{3,1};\mathbf{n}1^{1}_{3,1}\rangle\}$, $\{\langle\mathbf{F}^{1}_{3,2};\mathbf{C}^{1}_{3,2}\rangle,\langle\mathbf{P}^{1}_{3,2};\mathbf{M}^{1}_{3,2};\mathbf{n}1^{1}_{3,2}\rangle\}$, $\{\langle\mathbf{F}^{1}_{3,3};\mathbf{C}^{1}_{3,3}\rangle,\langle\mathbf{P}^{1}_{3,3};\mathbf{M}^{1}_{3,3};\mathbf{n}1^{1}_{3,3}\rangle\}$, $\{\langle\mathbf{F}^{1}_{3,4};\mathbf{C}^{1}_{3,4}\rangle,\langle\mathbf{P}^{1}_{3,4};\mathbf{M}^{1}_{3,4};\mathbf{n}1^{1}_{3,4}\rangle\}$, $\{\langle\mathbf{F}^{2}_{3,0};\mathbf{C}^{2}_{3,0}\rangle,\langle\mathbf{P}^{2}_{3,0};\mathbf{M}^{2}_{3,0};\mathbf{n}1^{2}_{3,0}\rangle\}$, $\{\langle\mathbf{F}^{2}_{3,1};\mathbf{C}^{2}_{3,1}\rangle,\langle\mathbf{P}^{2}_{3,1};\mathbf{M}^{2}_{3,1};\mathbf{n}1^{2}_{3,1}\rangle\}$, $\{\langle\mathbf{F}^{2}_{3,2};\mathbf{C}^{2}_{3,2}\rangle,\langle\mathbf{P}^{2}_{3,2};\mathbf{M}^{2}_{3,2};\mathbf{n}1^{2}_{3,2}\rangle\}$, $\{\langle\mathbf{F}^{2}_{3,3};\mathbf{C}^{2}_{3,3}\rangle,\langle\mathbf{P}^{2}_{3,3};\mathbf{M}^{2}_{3,3};\mathbf{n}1^{2}_{3,3}\rangle\}$, and $\{\langle\mathbf{F}^{2}_{3,4};\mathbf{C}^{2}_{3,4}\rangle,\langle\mathbf{P}^{2}_{3,4};\mathbf{M}^{2}_{3,4};\mathbf{n}1^{2}_{3,4}\rangle\}$. Columns 1–5 and 6–10 show data without and with Gaussian white noise, respectively. Rows 1–5 are the data obtained by rotations of 0°, 45°, 90°, 135°, and 180°. In addition, we select five common-shot gathers to create the augmented verification data set pairs using the same method by which the augmented training data set pairs are created.

The GT data set pairs and augmented training data set pairs are, respectively, sent to the DNN for training. The initial learning rate of Adam is 1×10^{-4}. The loss function in equation 15 is used to calculate the approximation degree between the estimated value obtained by the DNN and the ground truth for each training epoch. The closer these values are, the smaller MAE($\boldsymbol{\theta}$) is. This experiment has trained 500 epochs. We save the DNN parameters obtained from each epoch of training and select those that correspond to the minimum error of the verification set. When using the GT set, each epoch takes 4 s, and the obtained DNN parameters are $\boldsymbol{\theta}^{A}$. When using the augmented training data set, each epoch takes 179 s, and the obtained DNN parameters are $\boldsymbol{\theta}^{B}$. The iterative curves for the error loss function obtained from the general and augmented training data sets are shown in Figure 5. After convergence, the loss values of the training and verification sets obtained by the DNNGT are 3.7×10^{-4} and 4.5×10^{-4}, respectively, whereas the loss values obtained by the proposed DNNDAT are smaller, at 2.0×10^{-4} and 1.8×10^{-4}, respectively.

We use data that are not involved in the training stage to test the multiple suppression effect and antinoise ability (robustness) of the DNN. Figure 6a shows the clean full wavefield with multiples. To test the antinoise ability, varying intensity background noise is added to Figure 6a to obtain the noisy full wavefield with a signal-to-noise ratio (S/N) of 2 dB (Figure 6b). Here, the primaries and multiples are overwhelmed by background noise. Figure 6c shows the true surface-related multiples and Figure 6d shows the true primaries. Figure 7a shows the primary result obtained using the SRME method to suppress multiples in the clean full-wavefield data. The difference (Figure 7b) between the suppression result and true primaries shows that the multiple residues are relatively larger, and the prediction subtraction method attempts to maintain the balance between overfitting and the underfitting.

The data from Figure 6a and 6b are input into the DNN with the parameters $\boldsymbol{\theta}^{A}$ using GT, with the results shown in Figure 8. Figure 8a, 8e, and 8i shows the primaries reconstructed from the clean full wavefield using modes **P**-GT1, **P**-GT2, and **P**-GT3, respectively. Figure 8b, 8f, and 8j shows the differences between true primaries and Figure 8a, 8e, and 8i, respectively. Figure 8c, 8g, and 8k shows the primaries reconstructed from the noisy full wavefield using modes **P**-GT1, **P**-GT2, and **P**-GT3, respectively. Figure 8d, 8h, and 8l shows the differences between true primaries and Figure 8c, 8g, and 8k, respectively.

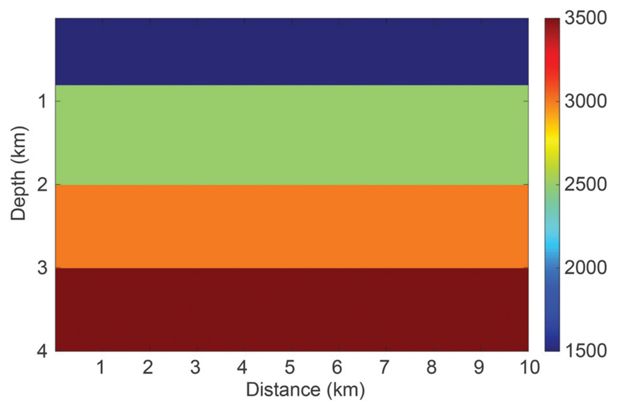

Figure 3. Simple velocity model with horizontal layers.

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Figure 8a, 8e, and 8i shows that the primaries obtained by modes **P**-GT1, **P**-GT2, and **P**-GT3 are almost completely reconstructed, proving that the DNNGT method can effectively suppress the multiples in the clean full wavefield. In Figure 8b, 8f, and 8j, only a small part of the primaries leak, and almost no residual multiples can be found. The position and intensity of the leaked primaries shown in Figure 8b and 8f are different, and they are averaged in Figure 8j, indicating that modes **P**-GT1, **P**-GT2, and **P**-GT3 can achieve good multiple suppression effects for the clean full wavefield. It should be noted that **P**-GT3 can combine the reconstruction effects of **P**-GT1 and **P**-GT2. The primaries shown in Figure 8c, 8g, and 8k are partially reconstructed; however, a large amount of background

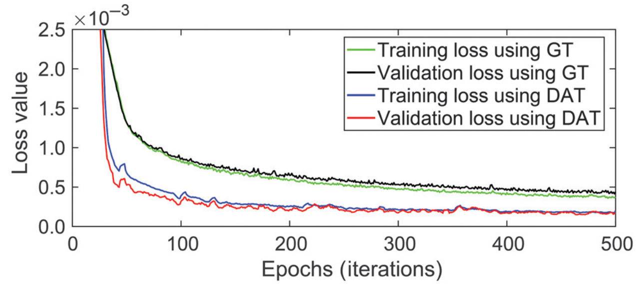

Figure 5. Training and verification loss curves for the first simple model.

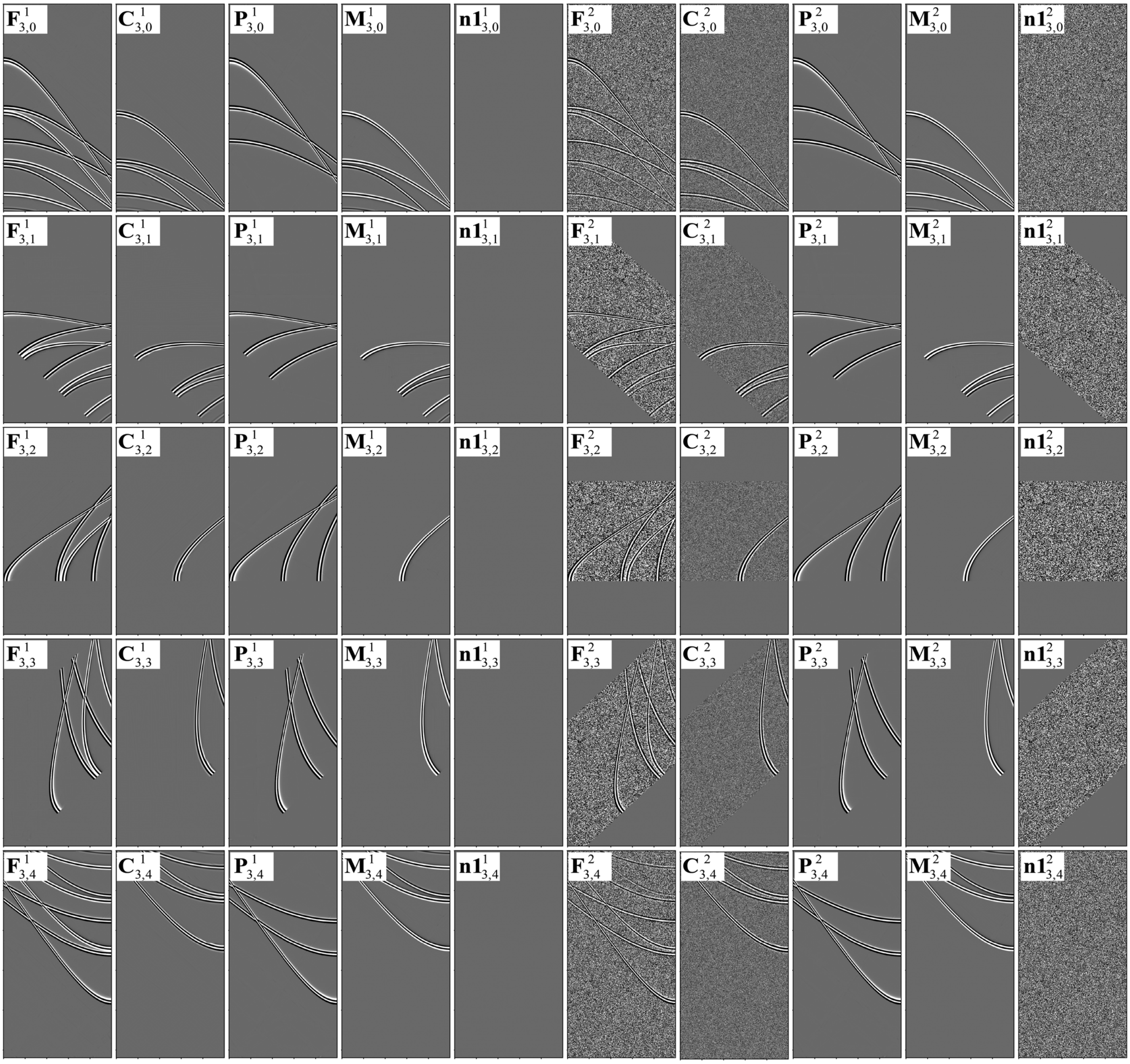

Figure 4. Partially augmented training data set pairs of the simple model. Columns 1–5 and 6–10 show data without and with Gaussian white noise, respectively. Rows 1–5 are the data obtained by rotation of 0°, 45°, 90°, 135°, and 180°.

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noise remains. From the differences shown in Figure 8d, 8h, and 8l, we identify a large number of primary leakage and background noise residues. The DNN with parameters $\boldsymbol{\theta}^A$ cannot reconstruct the primaries in the noisy full wavefield when background noise is present, which proves that DNNGT has poor antinoise stability.

We input the data from Figure 6a and 6b into the DNN with the $\boldsymbol{\theta}^B$ parameters using DAT. The reconstructed results using equations 21–23 are shown in Figure 9. Figure 9a, 9e, and 9i shows the primaries reconstructed from the clean full wavefield using modes **P**-DAT1, **P**-DAT2, and **P**-DAT3, respectively. Figure 9b, 9f, and 9j shows the differences between true primaries and Figure 9a, 9e, and 9i, respectively. Figure 9c, 9g, and 9k shows the primaries reconstructed from noisy full wavefield using modes **P**-DAT1, **P**-DAT2, and **P**-DAT3, respectively. Figure 9d, 9h, and 9l shows the differences between true primaries and Figure 9c, 9g, and 9k, respectively. In the primary reconstruction results of Figure 9a, 9c, 9e, 9g, 9i, and 9k, the primaries are almost completely reconstructed, and the separation effects are equivalent, without the residues from multiples and background noise. This demonstrates that the DNNDAT method has a good multiple suppression effect and good antinoise stability. The multiple suppression effects of modes **P**-DAT1, **P**-DAT2, and **P**-DAT3 are comparable. From the differences demonstrated in Figure 9b, 9d, 9f, 9h, 9j, and 9l, there is minimal leakage of the primaries, almost no residual from background noise or multiples, and no introduction of new noise.

Comparing the differences between Figures 8 and 9, the DNNDAT method shows less leakage of primaries when multiples in the clean full-wavefield data are suppressed. When suppressing multiples in noisy full-wavefield data, the DNNDAT method suppresses background noise more effectively than the DNNGT method. In addition, border effects are observed in Figure 8; however, they are not present in Figure 9. These border effects are the leakage of primaries and residues of multiples. This phenomenon occurs by rotating and adding background noise to increase the amount of training data set and improve the distribution of data during DNN training using the DAT method, which improves the robustness of the DNN and enables the identification of the edge features in full-wavefield data. Therefore, a better multiple suppression effect can be obtained along data edges when using the DNNDAT method. In contrast, owing to insufficient data with different distributions, there is a considerable border effect in the multiple suppression results obtained using the DNNGT method (Figure 8). The preceding comparison proves that the proposed DNNDAT method has a better multiple suppression effect and superior antinoise stability. The corresponding three calculation modes **P**-DAT1, **P**-DAT2, and **P**-DAT3 have a similar effect on the reconstruction of primaries. In addition, there are three modes to obtain the primary reconstruction results (i.e., equations 21–23), from which we can apply the mode with the best processing effect according to the actual results.

To quantitatively calculate the multiple suppression effect of the DNN, the primary reconstruction rate is defined as

$$R = 1 - \frac{|\mathbf{T}^{\mathrm{est}} - \mathbf{T}^{\mathrm{true}}|}{|\mathbf{T}^{\mathrm{true}}|} \times 100\%. \tag{24}$$

In equation 24, $\mathbf{T}^{\mathrm{true}}$ represents the true primaries and $\mathbf{T}^{\mathrm{est}}$ represents the reconstructed primaries. The higher the value of R, whose upper limit is one, the less the residual energy is present in the difference map, which means that the method causes less damage to the primaries and better suppression of multiples. Compared with the simple division of the reconstructed multiples and true multiples (such as the calculation of the separation S/N), R reflects the true suppression result. According to equation 24, the clean and noisy full wavefields with S/N ratios of 2 dB are processed using the DNN with parameters $\boldsymbol{\theta}^A$ to obtain Rs of 96.01% and −5.78%, respectively, for mode

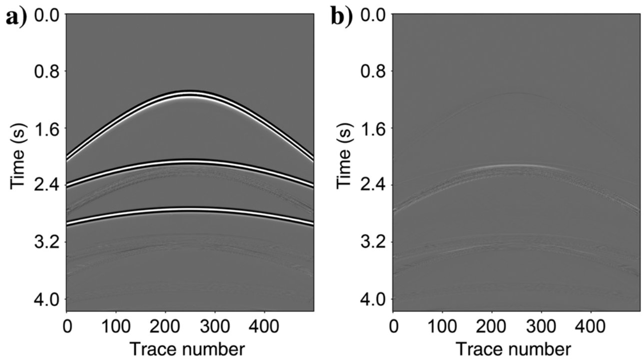

Figure 7. (a) Suppression results and (b) the difference between (a) and the true primaries using the SRME method to suppress multiples in a clean full wavefield.

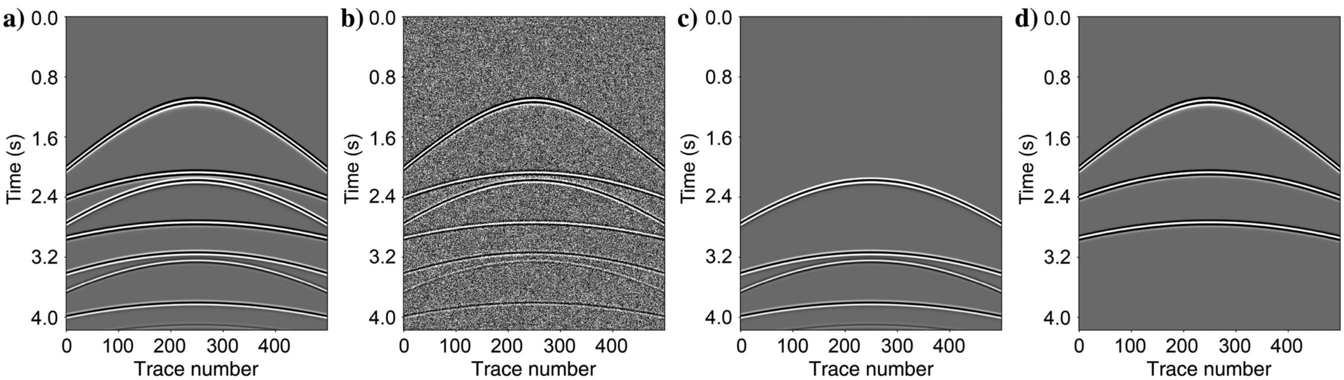

Figure 6. Test data of the first simple model. (a) Clean full wavefield with multiples, (b) noisy full wavefield with an S/N of 2 dB after adding background noise to (a), (c) true surface-related multiples, and (d) true primaries.

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P-GT1, 96.67% and −294.50%, respectively, for mode **P**-GT2, and 97.25% and −114.83%, respectively, for mode **P**-GT3. The clean and noisy full wavefields also are processed by the DNN with parameters $\boldsymbol{\theta}^B$ to obtain Rs of 98.58% and 97.11%, respectively, for mode **P**-DAT1, 98.40% and 95.94%, respectively, for mode **P**-DAT2, and 98.69% and 96.88%, respectively, for mode **P**-DAT3. When processing the clean full wavefield, the DNNGT and the DNNDAT can obtain good primary reconstruction results with very high R rates. Increases in background noise cause an increase in the residual multiples in DNNGT results, while weakening and decreasing mallssionsion effect. This indicates that the DNNGT is not suitable for the reconstruction of primaries from noisy full wavefields. However, under strong background noise, DNNDAT effectively reconstructs primaries and suppresses multiples. This also demonstrates that DAT improves the robustness of the DNNDAT.

Figure 10 shows the frequency spectrum for one trace of the training data, test data, and reconstruction results obtained using the DNNDAT. The frequency distribution range of the training data without background noise is consistent with that of the test data without background noise, and no artifacts are introduced in the frequency domain. Figure 10b and 10c shows that the frequency spectrum obtained by reconstructing the primaries in the clean and noisy full wavefields almost completely coincides with the frequency spectrum of the true primaries. This further supports the argument that the proposed DNNDAT method has good multiple suppression capability.

To understand the working process of the DNN, the test data shown in Figure 6b and the corresponding predicted multiples are used as inputs for the DNNDAT with $\boldsymbol{\theta}^B$. Figure 11 shows the feature maps of the partial convolutional layer outputs, which are used to visualize the intermediate activation. Figure 11a–11f

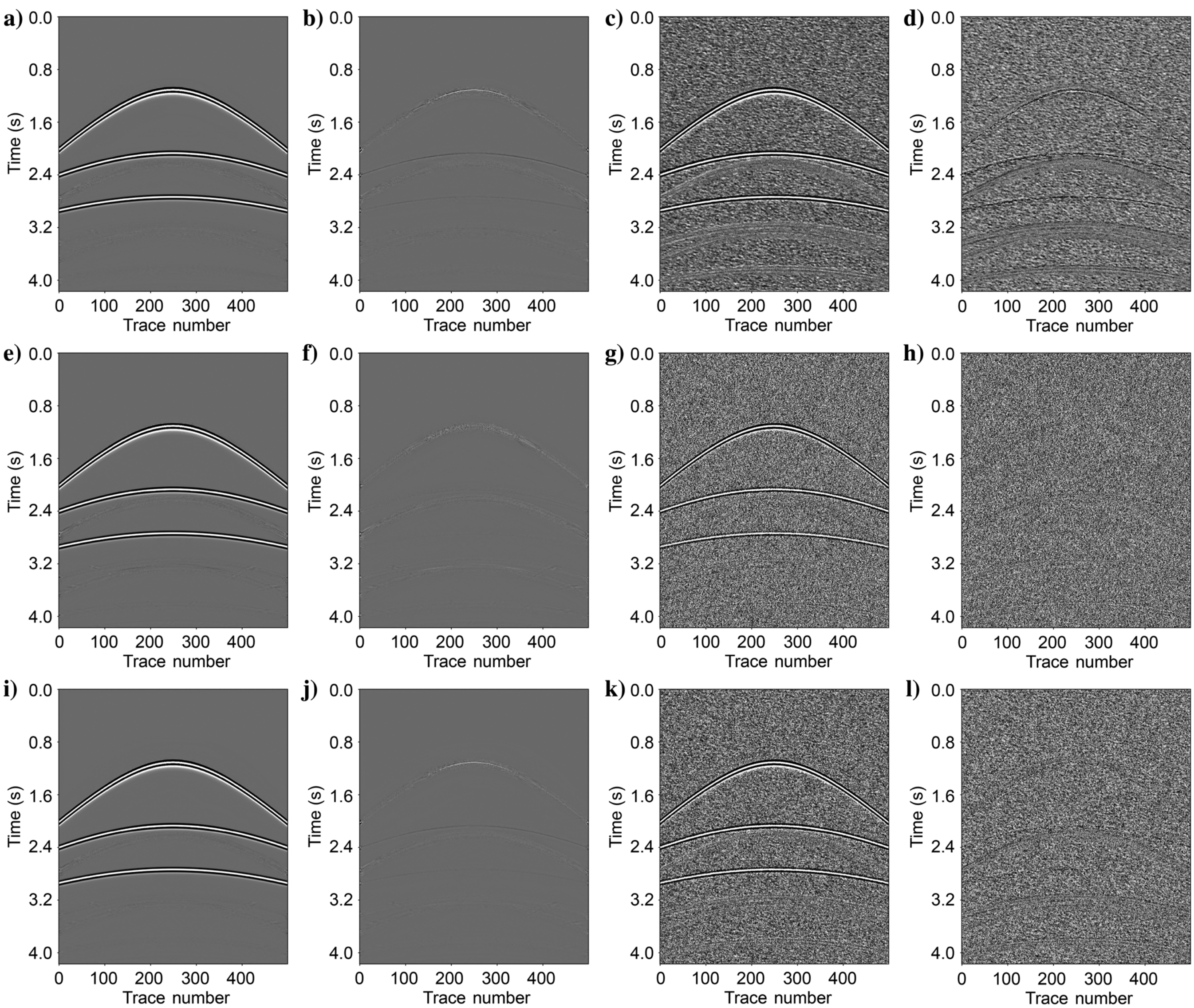

Figure 8. Reconstructed results and the corresponding differences using DNNGT. (a, e, and i) The primaries reconstructed from a clean full wavefield using modes **P**-GT1, **P**-GT2, and **P**-GT3, respectively. (b, f, and j) The differences between true primaries and (a, e, and i), respectively. (c, g, and k) The primaries reconstructed from a noisy full wavefield using modes **P**-GT1, **P**-GT2, and **P**-GT3, respectively. (d, h, and l) The differences between true primaries and (c, g, and k), respectively.

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shows the output values obtained from some channels for the convolutional layers V1–V6 shown in Figure 1. During convolutional encoding (Figure 11a), feature maps of the shallow layer retain all of the information in the noisy full-wavefield data. As the DNN depth increases (Figure 11c), the features of the primaries, multiples, and background noise gradually separates and the activation becomes increasingly abstract. Most of the feature maps shown in Figure 11e are blank, and the characteristics of primaries, multiples, and background noise are sparsely separated. During convolutional decoding, the sparse features (Figure 11e) of primaries learned

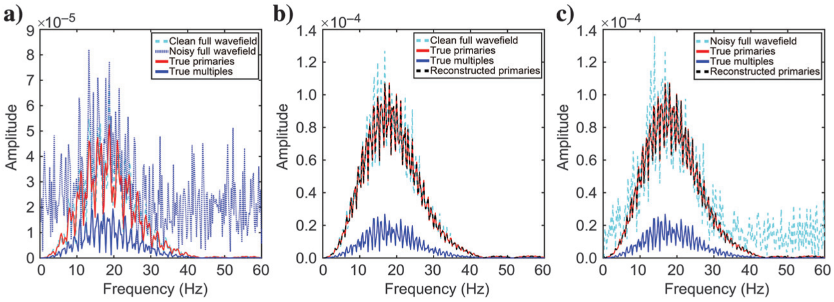

Figure 10. The frequency spectrum of one trace for training data and test data. (a) The frequency spectrum of the 251st traces for training data in the first row of Figure 4. (b) The frequency spectrum of the 251st traces shown in Figures 6 and 9i. (c) The frequency spectrum of the 251st traces shown in Figures 6 and 9k.

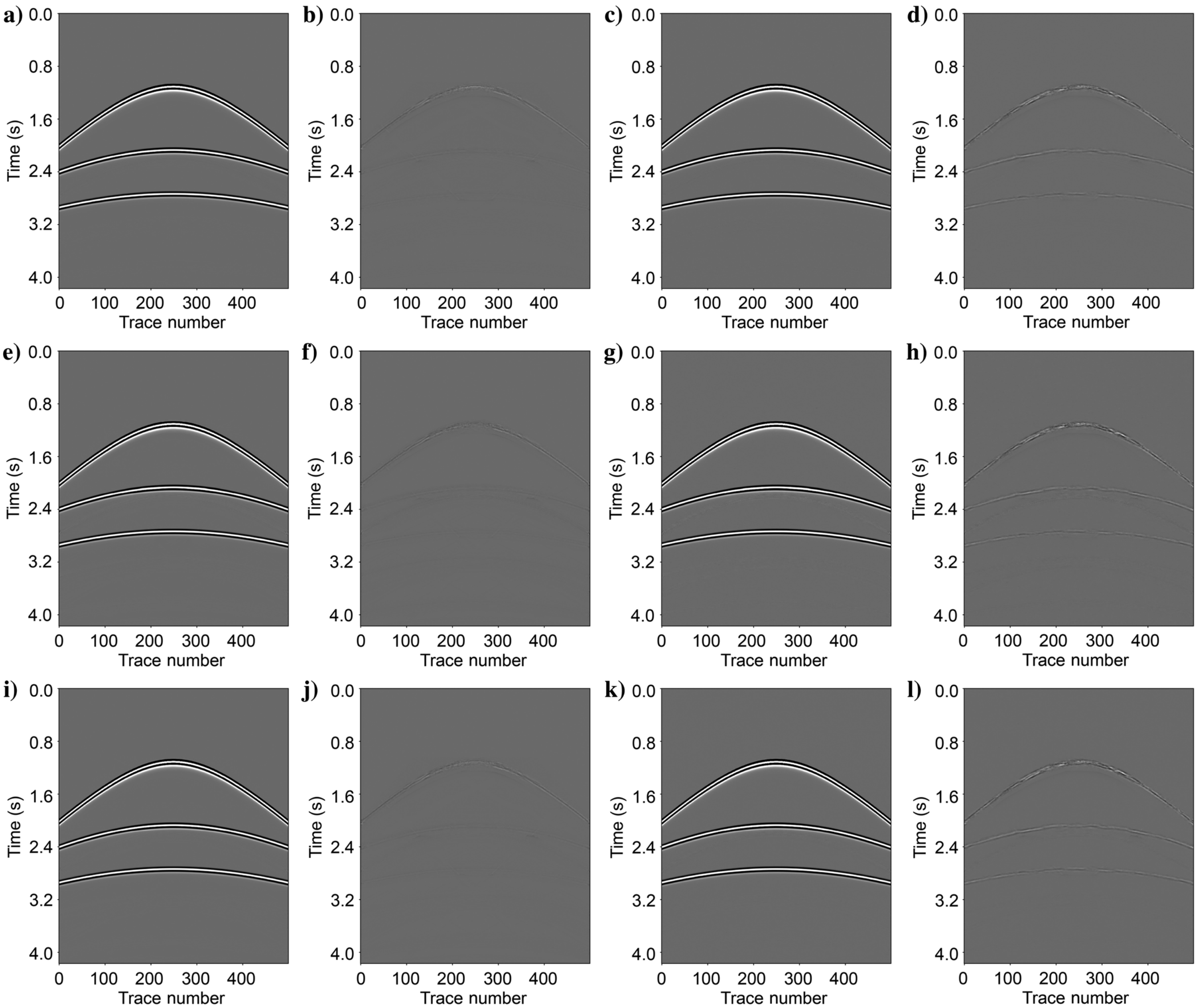

Figure 9. Reconstructed results and the corresponding differences using DNNDAT. (a, e, and i) The primaries reconstructed from clean full wavefield using modes **P**-DAT1, **P**-DAT2, and **P**-DAT3, respectively. (b, f, and j) The differences between true primaries and (a, e, and i), respectively. (c, g, and k) The primaries reconstructed from noisy full wavefield using modes **P**-DAT1, **P**-DAT2, and **P**-DAT3, respectively. (d, h, and l) The differences between true primaries and (c, g, and k), respectively.

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during the encoding process are gradually decoded and restored to a high-dimensional space, and the feature maps gradually become denser (Figure 11f). As the DNN depth increases (Figure 11d), the primaries and multiples gradually recover under their respective characteristics. In the almost deepest part of the DNN (Figure 11b), the primaries are almost completely reconstructed in some feature maps.

The data shown in Figure 6b and the corresponding predicted multiples are input into the DNN with parameters $\boldsymbol{\theta}^A$ trained by the GT to obtain feature maps (Figure 12). These maps show partial convolutional layers output in the DNN. In comparison to Figure 11, Figure 12 contains more noise, which has obscured features representing primaries and multiples from deeper DNN (Figure 12b and

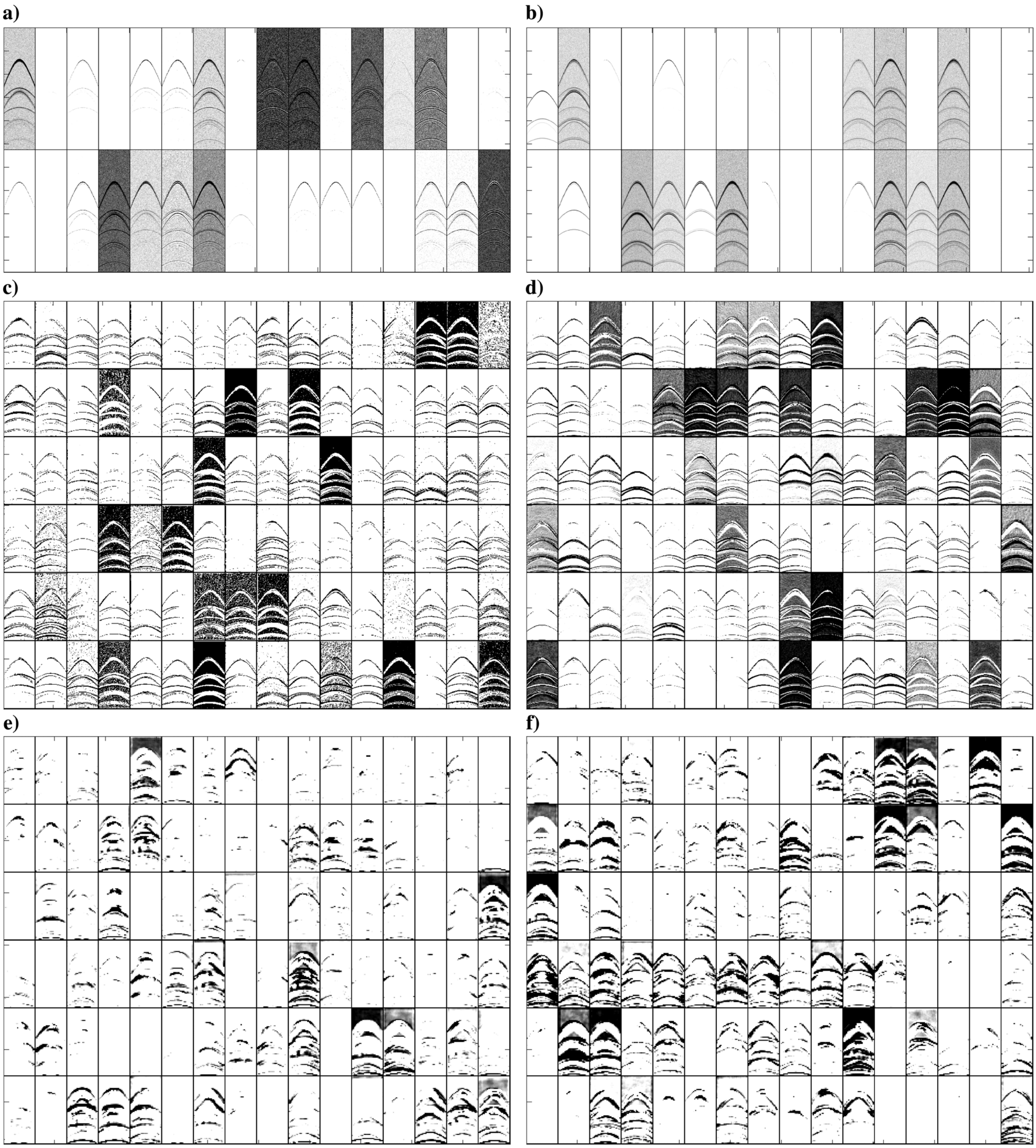

Figure 11. Visualization results. (a-f) Partial channels activated by the V1–V6 layers in the DNN trained by DAT.

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12d). This comparison further demonstrates that the DNNDAT method better distinguishes the characteristics of the primaries, multiples, and background noise in the noisy full-wavefield data than the DNNGT method.

Simple model two

The second simple velocity model is shown in Figure 13. This velocity model has four horizontal and one inclined interfaces to verify the generalization ability of the proposed method. Finite-difference

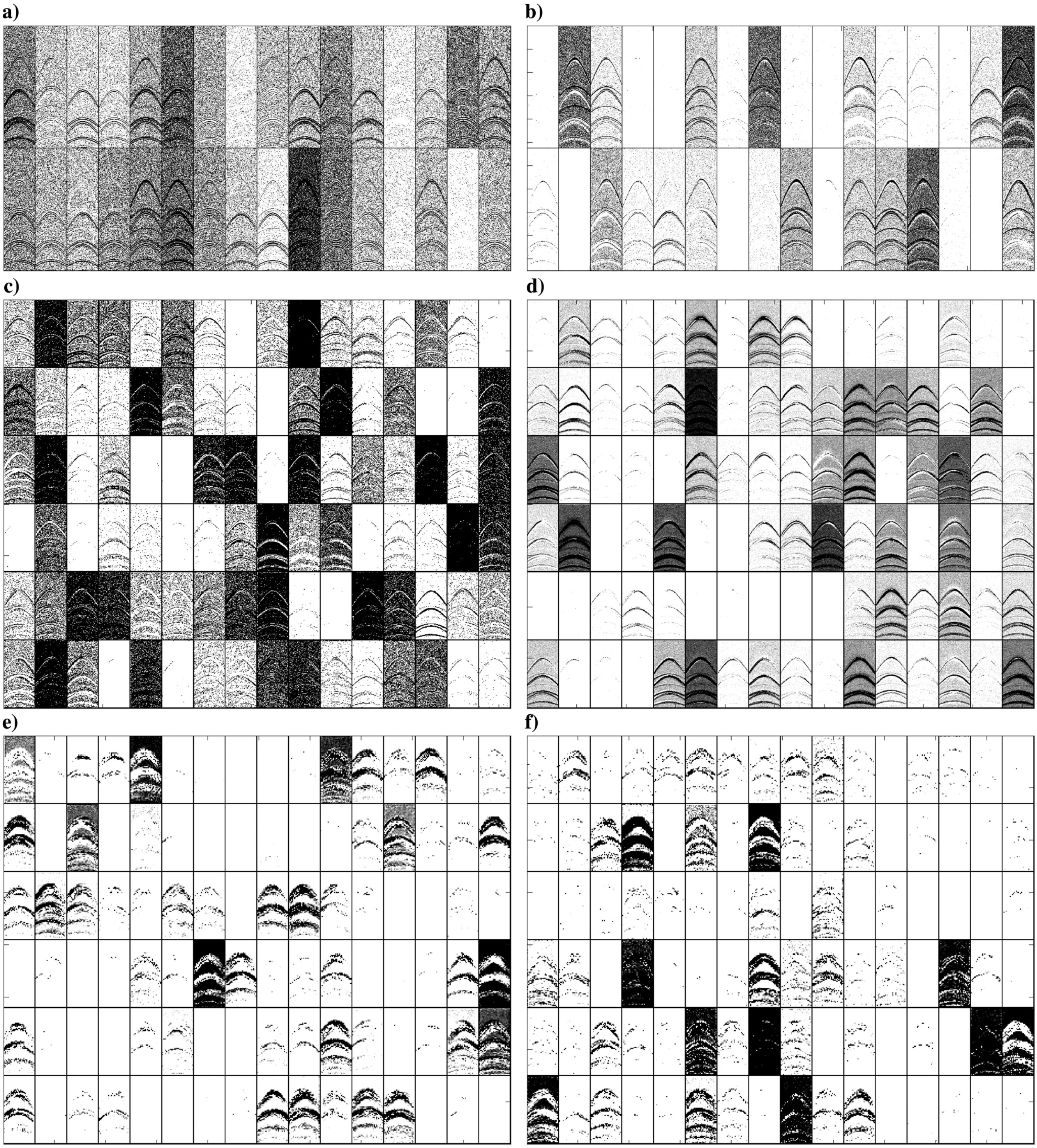

Figure 12. Visualization results. (a-f) Partial channels activated by the V1–V6 layers in the DNN trained by GT.

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forward modeling is used to obtain the common-shot gather of the full-wavefield data and the corresponding seismic data without surface-related multiples. Predicted multiples are obtained through convolution. In total, 400 receivers are used for each shot. The shot and receiver spacings are 10 m, and the sampling time interval is 4 ms.

During the training phase, we fix part of the DNN parameters $\boldsymbol{\theta}^B$ and improve the generalization ability of the DNN using the transfer learning method (Siahkoohi et al., 2019b; Wang et al., 2021). In total, 20 training sets are extracted for parameter training. First, the full-wavefield data are preprocessed using the SRME method to obtain label data for DNN training. Background noised is comprised of Gaussian white noise with 11 different amplitude intensities and is added to the full-wavefield data. Thereafter, the augmented training data set pairs are obtained by performing seven rotations at the 45° angles. Finally, the obtained 1760 augmented training data set pairs are sent to the DNN to train the unfixed parameters of the DNN and obtain the $\boldsymbol{\theta}^C$ parameters. In total, 100 training epochs are performed, with each epoch lasting 149 s.

The data shown in Figure 14 are used to test the multiple suppression effects and robustness capabilities of the proposed method. Figure 14a shows the clean full-wavefield data, Figure 14b shows the noisy full-wavefield data with an S/N of 6 dB, Figure 14c shows the true multiples, and Figure 14d shows the true primaries. It should be noted that the surface-related multiples shown in Figure 14b are particularly developed, and the background noise is strong. The results after surface-related multiples (from Figure 14a) are suppressed by the SRME method are shown in Figure 15a. The primary reconstruction rate is 94.93%, and the difference between the suppression results and true primaries is shown in Figure 15b. The suppression results show that the main primaries are not affected by the SRME method, and a small number of residual multiples can be seen in the difference.

To verify the generalization ability of the proposed method, the clean and noisy full-wavefield data from Figure 14a and 14b, respectively, are input into the DNN after the transfer learning process. Figure 16a shows the results from suppressing multiples in clean full-wavefield data using the proposed DNNDAT method with transfer learning (DNNDATTL). The primary reconstruction rate is 95.57%, which is slightly higher than the reconstruction rate obtained by the SRME method. This demonstrates that the DNNDATTL can effectively suppress multiples. Protection of the primaries using the DNNDATTL also is slightly better than that using the SRME method. The difference between the reconstructed primaries obtained using the DNNDATTL and the true primaries is shown in Figure 16b. This difference figure shows that the residual multiples obtained by the DNNDATTL are less than those obtained by the SRME method. The DNNDATTL also separates noisy full-wavefield data with an S/N of 6 dB to obtain the primaries, as shown in Figure 16c. Here, large amounts of background noise are eliminated and the multiples are well suppressed, resulting in a primary reconstruction rate of 93.74%. The primary reconstruction rate is

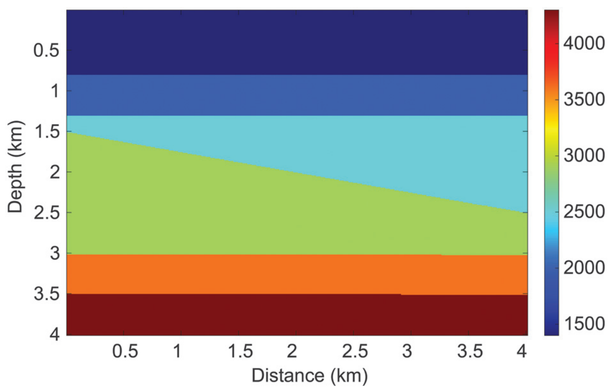

Figure 13. Second simple velocity model.

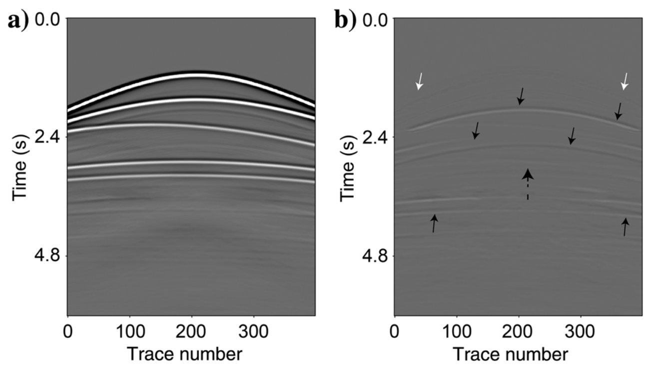

Figure 15. (a) Suppression results and (b) the difference between the true primaries and (a) using the SRME method to suppress multiples in clean full-wavefield data.

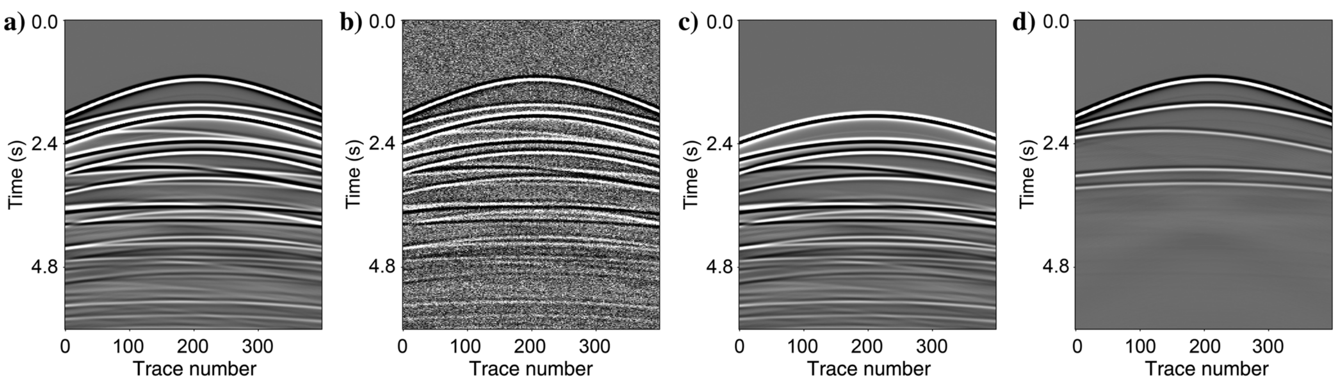

Figure 14. Test data of the second simple model. (a) Clean full-wavefield data, (b) noisy full-wavefield data, (c) true multiples, and (d) true primaries.

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comparable to that obtained using the SRME method and by suppressing multiples in the clean full-wavefield data using the DNNDATTL. This further demonstrates the superior antinoise stability of the proposed method. Figure 16d shows the difference between the true primaries and the data from Figure 16c and shows that there is almost no residual background noise. By comparing Figures 15 and 16, more residual multiples can be observed in data from the SRME method (the solid black arrows). The results obtained from the DNNDATTL processing of the clean full-wavefield data cause a larger primary leakage at the dashed black arrow shown in Figure 16b. The DNNDATTL processing of the noisy full-wavefield data results in a larger leakage of primaries at the white arrows shown in Figure 16d. The preceding comparison further proves that the DNNDATTL has good antinoise stability, and its ability to suppress multiples is slightly better than that of the SRME method.

Pluto model

The Pluto model is used to test the ability of the DNNDAT for reconstructing the primaries and suppressing the multiples and background noise. This model is known for having strong underwater surface multiples and top salt dome multiples. It is a complex geologic model that is used to test methods of multiple suppression in the petroleum industry. During simulation, 336 shots and receivers are placed horizontally along the surface of the model, at 30 m spacings. The sampling interval is 8 ms.

In the DNN training stage, 60 common-shot gathers are selected. To increase the volume of the training set and the robustness and generalization of the DNN, using equations 9 and 10, we add six different intensities of Gaussian white noise to each input data. These are then rotated seven times at a 45° angle to obtain 2880 augmented training data set pairs. In addition, we select five common-shot gathers to add 20 types of background noise with different intensities and perform the same rotation to create a set of augmented verification pair. The experiment trains 400 epochs, with 312 s for each epoch. When training is completed, the DNN parameters are recorded as $\boldsymbol{\theta}^D$.

To test the ability of primary reconstruction and antinoise stability using the DNNDAT, the clean and noisy full-wavefield data without being used for training the DNN are directly input into the DNN with parameters $\boldsymbol{\theta}^D$ after amplitude normalization. Figure 17a shows the clean full wavefield including multiples. After adding Gaussian white noise to Figure 17a, a noisy full wavefield with an S/N of 5 dB is obtained (Figure 17b), which shows that background noise is

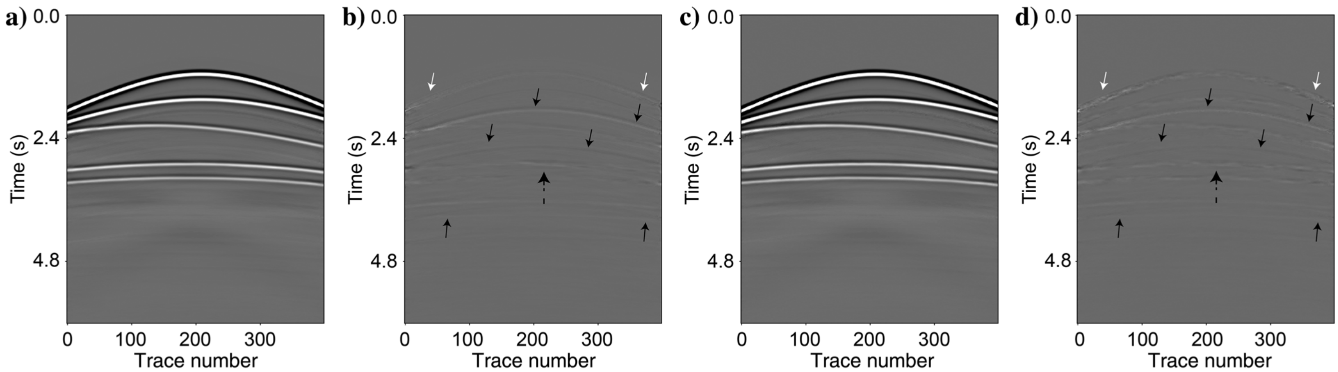

Figure 16. Results and differences using the DNNDATTL with mode **P**-DAT3 to suppress multiples. (a) Results suppressed from clean full-wavefield data. (b) Difference between the true primaries and (a). (c) Results suppressed from the noisy full-wavefield data. (d) Difference between the true primaries and (c).

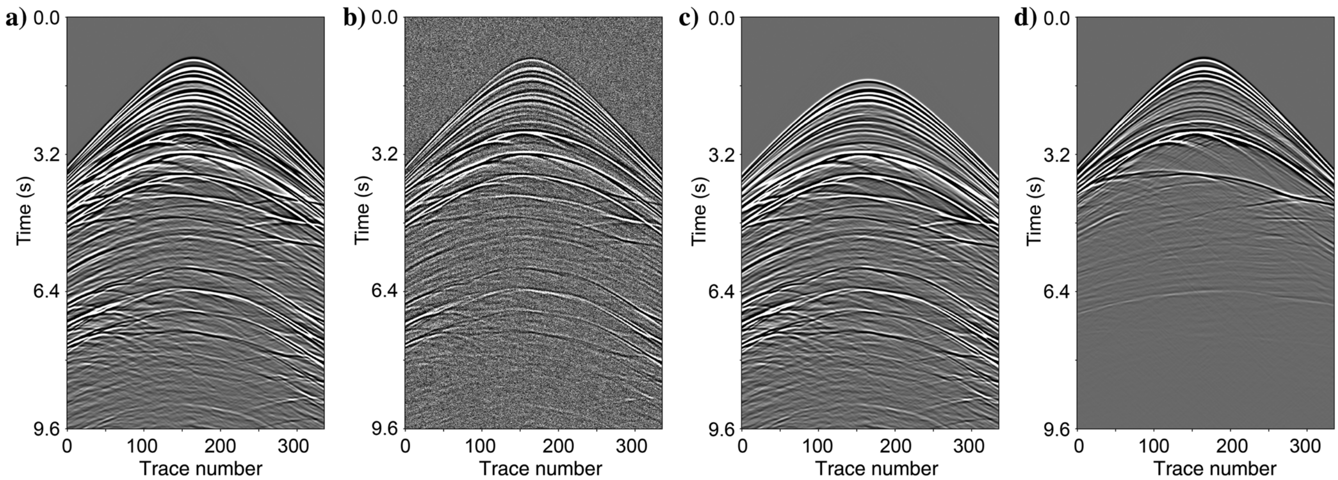

Figure 17. The 165th common-shot gather. (a) Clean full wavefield, (b) noisy full wavefield, (c) true multiples, and (d) true primaries.

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dispersed throughout the seismic data. Figure 17c shows the true multiples and Figure 17d shows the true primaries. The primaries are mainly distributed in the gather recorded before 6.4 s, and the multiples are mainly distributed in the gather recorded after 1.6 s. The multiples considerably interfere with the primaries.

The data shown in Figure 17a and 17b are input into the DNN with parameters $\boldsymbol{\theta}^D$, and the reconstruction results of the primaries using the DNNDAT are shown in Figure 18. Figure 18a and 18c shows the primaries reconstructed by the proposed method, and the differences between them and the true primaries are shown in Figure 18b and 18d. The primaries in the clean and noisy full wavefields are almost completely reconstructed, and there is almost no residual background noise. By comparison, the primaries are almost completely reconstructed correctly with only a small amount of leakage at depth. Using equation 24, the Rs using the DNNDAT to process the clean and noisy full wavefields is 93.53% and 91.95%, respectively. Therefore, there is a good suppression effect of the multiples. The proposed DNNDAT method simultaneously suppresses multiples and background noise, whereas restoration and reconstruction of the primaries occur without introducing new noise.

Common-offset sections can be used to better observe the effect of the primary reconstruction and multiple suppression. Figure 19a shows the common zero-offset section from the clean full wavefield with multiples. Apparent surface-related multiples can be seen approximately 3.2 and 6.4 s. The common-offset section from the noisy full wavefield after adding background noise to the clean full wavefield is shown in Figure 19b. Here, the background noise is dispersed throughout the profile, resulting in the complete loss of some primaries and multiples. A common-offset section from the true primary is shown in Figure 19c. The deep weak signal shown by the black arrow is almost completely lost due to interference from multiples and background noise. The full-wavefield data with two different background noises are processed by DNNDAT to obtain the common zero-offset sections (Figure 20a and 20b). Here, the primaries are recovered well, and the multiples and background noise are fully suppressed. The deep signal indicated by the black arrow is almost completely restored. To better analyze the suppression effect of the multiples and background noise, the differences between the true and reconstructed primary sections are shown in Figure 20c and 20d, respectively.

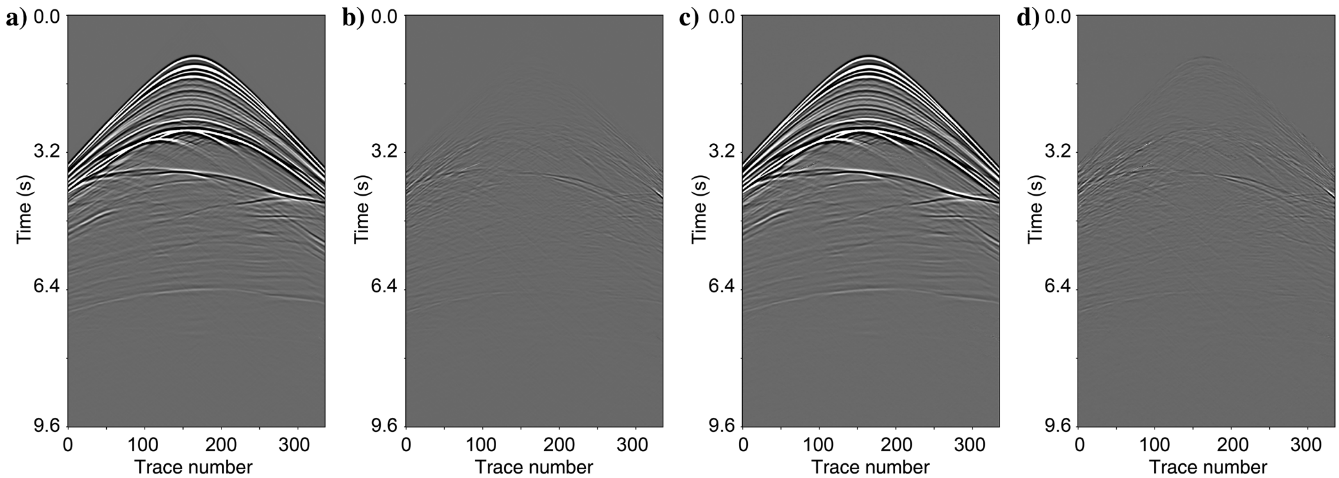

Figure 18. The reconstruction results using the DNNDAT. (a and c) The primaries reconstructed from the clean and noisy full wavefields, respectively. (b and d) The differences between the true primaries and (a and c), respectively.

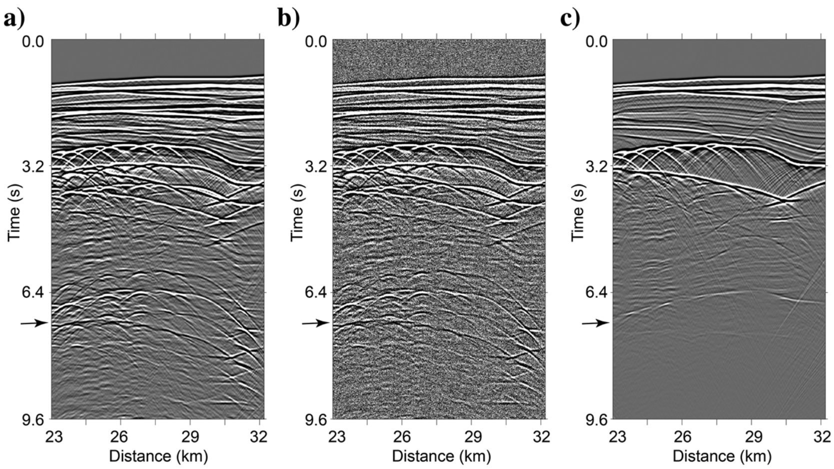

Figure 19. Common zero-offset section from (a) clean full wavefield, (b) noisy full wavefield, and (c) true primaries.

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It can be seen that a large amount of background noise remains in Figure 19b. However, no background noise pollution or new noise is introduced in Figure 20a and 20b, in which there is a similar amount of primary leakage. For the Pluto model, the proposed DNNDAT method effectively reconstructs primaries, recognizes and suppresses multiples, eliminates the background noise, and has a good robustness.

Field data

The 2D field data are collected along a 2D marine seismic line with 1079 shots, each containing 132 traces. The shot and receiver intervals are 16.67 m. Initially, real noise is extracted from marine data and used as background noise (Hu and White, 1998). Subsequently, 280 full-wavefield data are added with four different intensities of background noise. All data generated are rotated seven times at 45° intervals to obtain input data for training the DNN. The label data for training the DNN are obtained when the selected full-wavefield data are preprocessed using the Radon method by deleting the multiple signal coefficients and preserving the primary signal coefficients. In other words, the field data are preprocessed using a conventional multiple suppression method. The preprocessing time per shot is 186 s. In the training phase, the DNN is trained using parameters $\boldsymbol{\theta}^D$ through transfer learning to obtain parameters $\boldsymbol{\theta}^E$ with DNNDAT. Each epoch takes 796 s.

Figure 21a shows the original full-wavefield data for the DNN testing phase, in which a large number of multiples exist between 1.2 and 1.9 s. In the testing phase, background noise (Figure 21b) is added to the original full-wavefield data to obtain noisy full-wavefield data. The results obtained from suppressing multiples in Figure 21a using the Radon method are presented in Figure 22a. The results obtained by suppressing the multiples in Figure 21a and 21c with parameters $\boldsymbol{\theta}^E$ are shown in Figure 22b and 22c, respectively. There are more residual multiples in results obtained by the Radon method, as indicated by the solid black arrow in Figure 22a. The dashed white arrows shown in Figure 22b and 22c indicate less

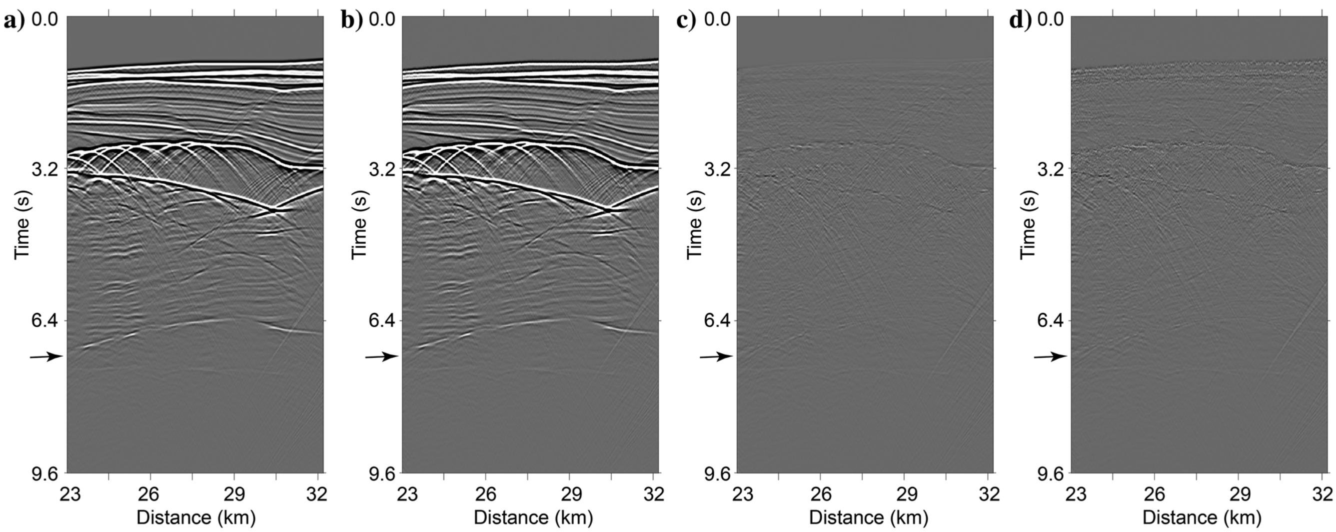
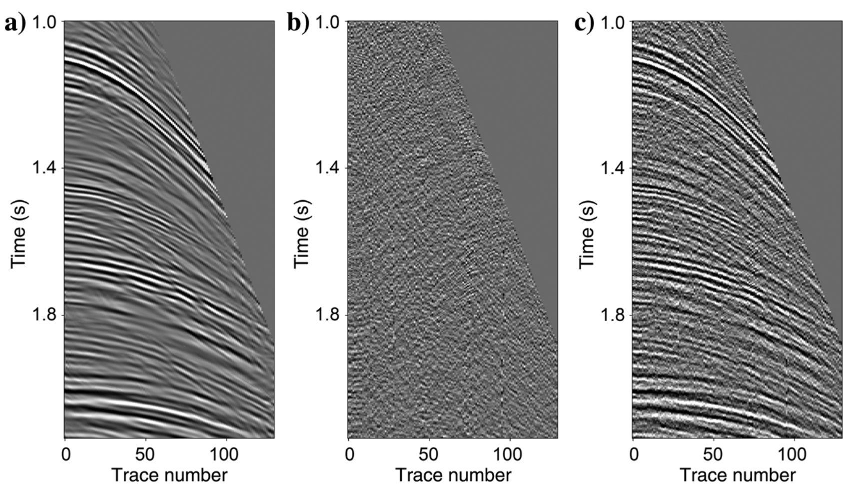


Figure 20. Common zero-offset section from reconstructed primaries using the DNNDAT to process (a) clean and (b) noisy full wavefields. (c and d) The differences between the common-offset section from true primaries and (a and b), respectively.

Figure 21. Full-wavefield data for the field test. (a) Data without background noise, (b) background noise, and (c) data after the addition of background noise.



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damage of primaries. In the DNNDAT method results, the energy of multiples recorded between 1.2 and 1.9 s is greatly reduced. The solid white arrows across Figure 22 show that the proposed DNNDAT method better suppresses multiples in the original full-wavefield data rather than that in the noisy full-wavefield data and the results of the proposed DNNDAT method also are better than those of the Radon method.

Figure 23a shows part of the nearest-offset section obtained from original full-wavefield data. The red arrows in the figure indicate strong energy multiples that are well developed. Figure 23b shows a section from the noisy full-wavefield data, in which background noise obscures part of the primary and multiple signals. Figure 23c shows the section obtained by suppressing multiples in the original full-wavefield data using the Radon method. Figure 23d and 23e shows the sections obtained by suppressing the multiples of the original and noisy full wavefields using the proposed DNNDAT method, respectively. The multiples at the red arrows shown in Figure 23c–23e are well suppressed. However, the Radon method causes more multiple residuals (the yellow arrows shown in Figure 23c) and more leakage of the primary signal (the blue arrows shown in Figure 23c). In Figure 23d and 23e, the effects of multiple suppression and primary protection are better. This further suggests that the proposed DNNDAT method achieves a good separation effect and has good robustness under transfer learning.

DISCUSSION

In the "Example" section, the common-shot gather data without multiples are required as label data to produce the training data. When the label data of each work area are insufficient, the traditional multiple suppression methods are required to obtain the label data. With the continuous accumulation of label data, we can use seismic data from different work areas to jointly train the DNN to further improve the generalization ability of the DNN.

The DNN structure designed in Figure 1 is similar to that of U-net, which can extract the characteristics of primaries and multiples. Encoding maps the seismic data to the sparse domain, which is used to extract data features. Decoding separates the characteristic coefficients of the primaries, multiples, and background noise, which are used to reconstruct the corresponding signals. Compared with U-net, the proposed DNN can more flexibly set the number of network layers used during encoding and decoding according to the available training data. It also can better reconstruct the primaries. ResNet is a short-circuit mechanism DNN that adds a residual unit, which solves the problem of gradient disappearance during the training process and is suitable for DNNs with large depths. However, when the seismic data set is large, ResNet occupies a larger graphic processing unit (GPU) memory, the training time increases, and it is more sensitive to fluctuations in the data. Therefore, we do not use ResNet to reconstruct the primaries. GAN is a generative model that is mainly used for image generation and data enhancement, such as improving image resolution and style transfer. It is composed of two parts: generators and discriminators, which are affected by common DNN structures. This means that the results for a GAN are closely related to the structure of the selected DNN. The proposed DNN can serve as a generator for GAN. However, because a GAN is difficult to train, it often incurs higher learning costs.

Network layers

To verify the impact of different network layers on primary reconstruction and multiple suppression, we use the DNN structure outlined in Figure 1. The network structure acts as the basis to increase or decrease the number of convolutional layers during convolutional encoding and decoding, respectively. This is then tested using the first simple model data. We reduce and increase the number of convolutional layers before and after the first three Concats, respectively. From this, we obtain a series of network structure net-A through to net-G, which has varying amounts of encoding and decoding layers in Table 1. We use the first simple model data to train and test the seven networks. In the test stage, the primary reconstruction rates obtained by reconstructing the primaries from clean and noisy full-wavefield data are shown in Figure 24. The primary reconstruction rates obtained by the net-A to net-E networks are very high, >0.9, with the net-C network having the

Table 1. Numbers of convolutional layers for seven network structures during convolutional encoding and decoding.

Network structure	Encoding layers	Decoding layers
Net-A	14	11
Net-B	11	14
Net-C	14	14
Net-D	11	17
Net-E	14	17
Net-F	17	17
Net-G	14	20

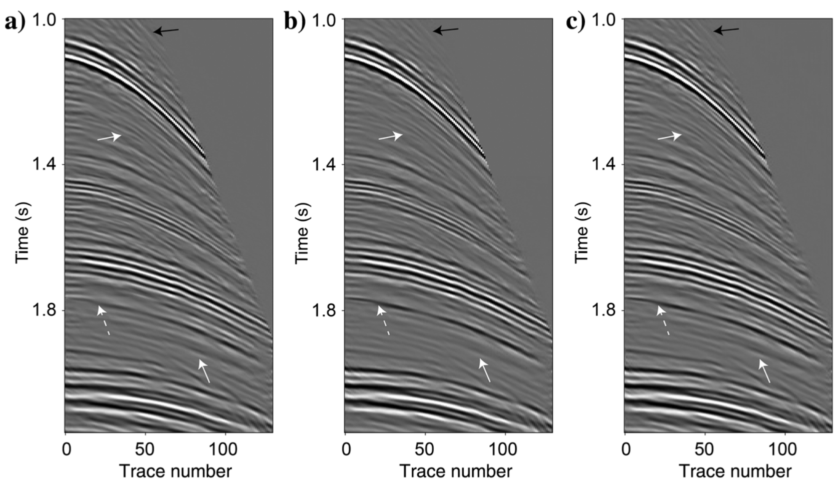

Figure 22. Reconstructed primaries after suppressing multiples in the original full wavefield using (a) the Radon method and (b) the proposed DNNDAT method. (c) The proposed DNNDAT method to suppress multiples in the noisy full wavefield with background noise.

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highest rate. These rates are much higher than those obtained by the net-F and net-G networks. Figure 25 shows the loss curves of the training and validation data sets for the different network layers. Here, the loss values of the training and validation data sets from net-F and net-G are much higher than those of the other networks, whereas the loss values of the training and verification data sets from the net-C network are the smallest. The loss curves from the net-G network have overfitting issues, which explains the low primary reconstruction rate seen in Figure 24. In general, more DNN layers provide sufficient parameters and improve the ability of the DNN to effectively suppress multiples. However, smaller training data sets are susceptible to overfitting when there are too many DNN layers, which agree with the findings of Yu et al. (2019). Figures 24 and 25 demonstrate that the network structure proposed in Figure 1 is reasonable.

Rotations in training data

We use the first simple data set to explore the effect of using rotation skills in the training data. We retrain the DNN using the same experimental conditions as the first simple data experiment; however, the rotation skill is not used in the data augmentation process. The loss curves of the training and verification data sets are shown in Figure 26. After training, the parameters $\boldsymbol{\theta}^F$ of the DNN are obtained using the DAT method without rotation. The final loss values of the training and verification data sets are 7.7×10^{-4} and 6.3×10^{-4}, respectively, which are much larger than the loss values obtained using the DAT method with rotation (Figure 5). The training method without rotation is difficult to converge and fluctuates greatly during the early epochs in Figure 26. Similar to mode **P**-DAT3, the DNN with parameters $\boldsymbol{\theta}^F$ reconstructs the primaries in Figure 6a and 6b, the results from which are shown in Figure 27. Here, the primary reconstruction rates are 97.63% and 88.45%. Compared with the separation result using the DNNDAT with rotation, there is more primary leakage and multiple residues present in the primary reconstruction results (Figure 27) and residual background noise (Figure 27c). This comparison shows that the lack of rotation in the DAT method can cause slow or no convergence during training, resulting in a poor primary reconstruction effect.

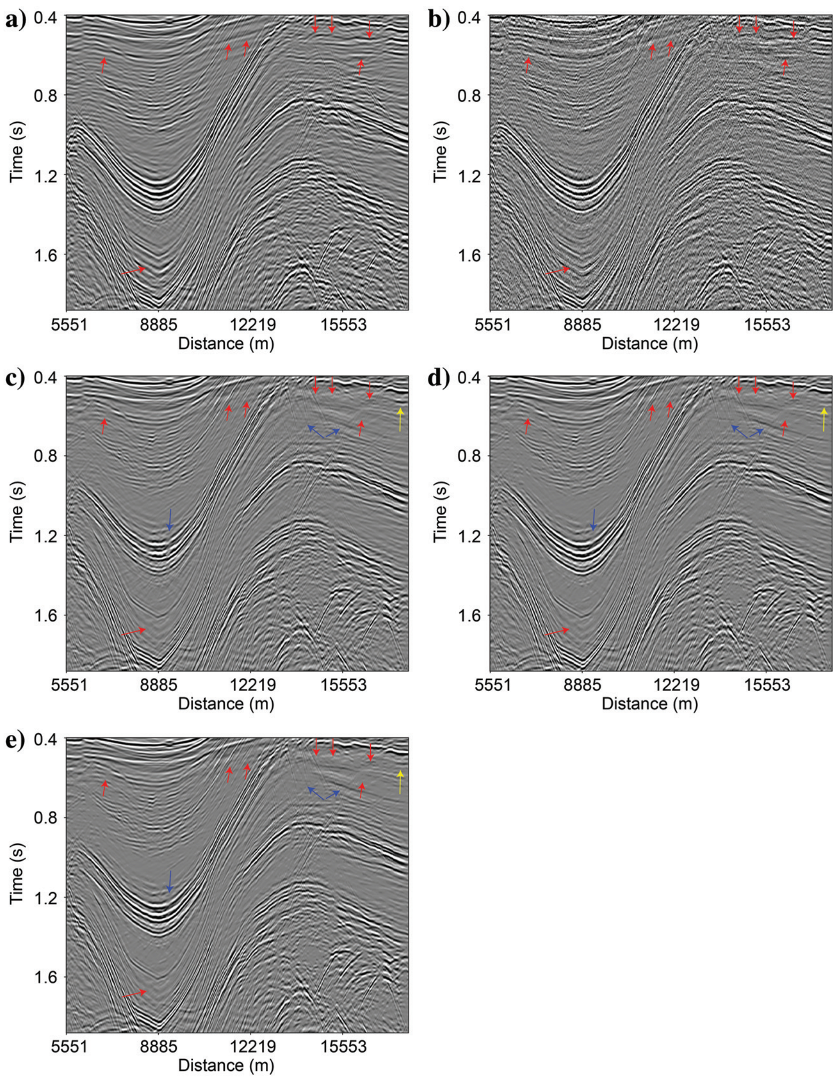

Figure 23. Nearest-offset sections before and after multiple suppression. (a and b) Sections of the original and noisy full wavefields, respectively. (c and d) Sections obtained by suppressing multiples in the original full wavefield using the Radon method and the proposed DNNDAT method, respectively. (e) Section obtained by suppressing multiples in the full wavefield with the background noise using the proposed DNNDAT method.

Role of the predicted multiples and transfer learning

To discuss the role of the predicted multiples, we use the first simple model data to train the DNN without adding predicted multiples to the input data and obtain the DNN parameters $\boldsymbol{\theta}^G$. Similar to mode **P**-DAT3, the DNN with parameters $\boldsymbol{\theta}^G$ is used to reconstruct the primaries in Figure 6a and 6b, and the results obtained are shown in Figure 28a and 28c, respectively. The primary reconstruction rates are 98.37% and 95.75%, respectively. The differences between Figure 28a and 28c and the true primaries are shown in Figure 28b and 28d, respectively. The DNN trained without adding the predicted multiples obtains good primary reconstruction results; however, the multiple suppression effect is slightly lower than that of the proposed DNNDAT method using the predicted multiples as a channel for input data. Figure 29 shows the feature maps of partial convolutional layers output for the experimental DNN trained without the predicted multiples. By comparing Figures 11 and 29, fewer feature maps representing true multiples exist in Figure 29. For example, Figure 11d has 12 feature maps; however, Figure 29d has five. The preceding comparison shows that adding predicted multiples to the training data set is helpful for identifying true multiples (Siahkoohi et al., 2019a) and then extracting their corresponding features.

The data shown in Figure 14a are fed into the DNN with the parameters $\boldsymbol{\theta}^G$ and $\boldsymbol{\theta}^B$. The pri-

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mary reconstruction results obtained without using transfer learning are shown in Figure 30a and 30c, respectively. The differences between the reconstructed primary results and true primaries are shown in Figure 30b and 30d, respectively. Residual multiples exist in Figure 30a and 30c, but the multiples shown in Figure 30c are smaller. Figure 30d difference plot shows that the leaked primaries are substantially less than those in Figure 30b. This comparison shows that, when multiples in seismic data from the same work area are suppressed, good separation results can be obtained by training the DNN with and without predicted multiples. However, when the multiples are suppressed in data from other work areas, the DNN trained with the predicted multiples is more effective than the DNN trained without the predicted multiples. The primary reconstruction result of the DNN under transfer learning is shown in Figure 16a, and it has a good primary reconstruction effect. However, without transfer learning, the primary reconstruction result of the DNN is poor (Figure 30c). Comparing Figures 16a and 30c further demonstrates the importance of transfer learning for DNNs in suppressing multiples across different work areas.

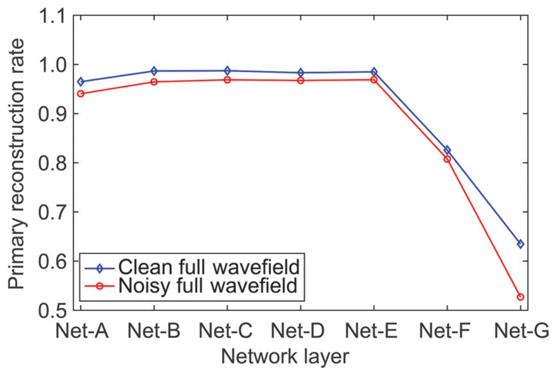

Figure 24. Primary reconstruction rates using different network layers. Net-A to net-G represent the networks with varying numbers of encoding and decoding layers. The blue and red lines show rates for the clean and noisy full wavefields, respectively.

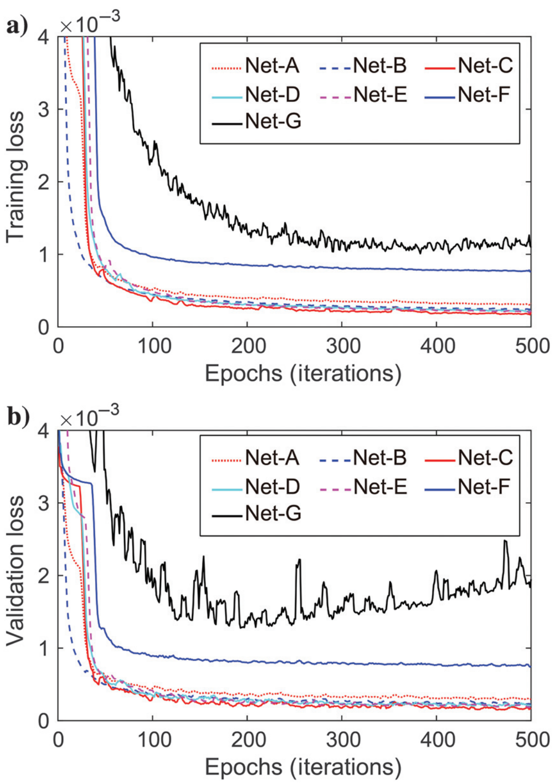

Figure 25. Loss curves of (a) training and (b) verification data sets under different network layers.

We want to determine whether transfer learning produces more accurate results on the field data compared with training from scratch using the synthetic data. To do this, we use the data sets of the first three models (simple model one and two, and the Pluto model) to train the DNN. The trained DNN is then used to suppress multiples in the field data from Figure 21a. The reconstructed primary results are shown in Figure 31. Compared with Figure 22b, the white arrow (Figure 31) indicates more residual multiples, whereas the black arrows (Figure 31) indicate larger leakage and damage to primary signals. This proves that transfer learning can produce more accurate results than training from scratch. There are some reasons for this phenomenon. Using only data from three velocity models to train the DNN prevents the DNN from being used to suppress multiples in other completely different velocity models. As such, transfer learning is a better choice when there are fewer types of data. Because the data in the first three examples are all synthetic, the distributions of these data are different from those of the field data. A DNN trained using synthetic data may not be efficient for suppressing multiples in field data. However, as the types of data from different velocity models increase, training from scratch using those data may obtain better results than the transfer learning method.

Background noise in the input data

Background noise is added to the input data to improve DNN effectiveness for suppressing multiples and protecting primaries by changing the data distribution. Adding background noise also enables the DNN to have antinoise stability. When the training data set is small, background noise often is often added to avoid overfitting and achieve better results from the DNN (Huang et al., 2020; Kim et al., 2021). Adding background noise also enables the DNN to learn more robust features and better suppress multiples. Considering the complexity of the geologic record, the primary and multiple models learned by the DNN are difficult to fully express the

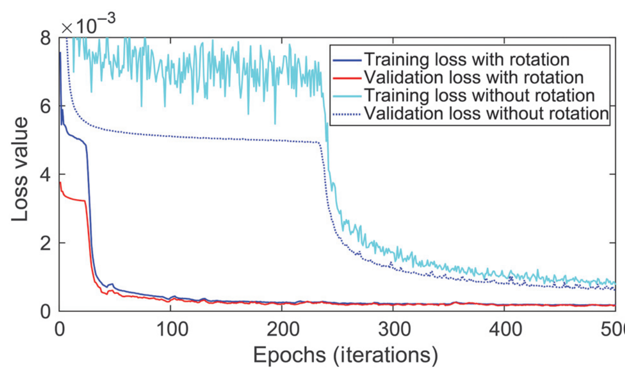

Figure 26. Loss curves of the training and verification data sets using the DNNDAT method with and without rotation.

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actual formation conditions, resulting in differences in the wavefield dynamics and kinematics. Therefore, to increase the robustness of the DNN and reduce the instability caused by the difference in field data, the addition of background noise to the training data set is essential.

To further verify this, we use the first simple model data to test the suppression effect of multiples using the DNNDAT method without adding background noise (DNNDATWABN) during the training and testing stages. When creating the training data set, we rotate the training data set without adding background noise to the input data. Equations 21–23 are applied to DNNDATWABN with the results shown in Figure 32. The calculated primary reconstruction rates are 97.72%, 97.81%, and 98.18% for equations 21–23, respectively. The GT method neither adds background noise nor rotates the input data, whereas the DAT method adds background noise and rotates the input data. Compared with the proposed DNNDAT method (comparison between Figures 9 and 32), the DNNDATWABN causes more primary leakage. The primary reconstruction rates obtained by the DNNDATWABN method are higher than those obtained by the DNNGT method, but lower than those obtained by the proposed DNNDAT method. The preceding test shows that adding background noise to the input data to train the DNN can improve the protection of primary signals and better suppress multiples.

Loss functions

The loss function is used to show the difference between the output results and the ground truth. Different loss functions determine different optimization strategies. In general, the mean squared error (MSE) loss function calculates the loss value by squaring the error. Therefore, the MSE magnifies outliers, resulting in a less robust model that is susceptible to outliers. The MAE loss function is more robust against outliers than MSE (Guitton and Verschuur, 2004; Jiang et al., 2020). However, the gradient of the MAE is consistent in most cases and remains large, which is not conducive to the model convergence. The Huber loss function (Guitton and Symes, 2003) combines the characteristics of MSE and MAE with the hyperparameter $\boldsymbol{\delta}$, which reduces the sensitivity to outliers and achieves a more uniform divergence. The structural similarity index (SSIM) loss function evaluates the difference between the two data through brightness, contrast, and structural formulas, and it makes use of the local features in the data (Zhou et al., 2004). We use MSE, SSIM, MAE, and Huber with different hyperparameters as loss functions to evaluate the effects of multiple suppression and primary reconstruction by the proposed DNNDAT. We repeat the previous experiments using the first and second simple model data, however, with different loss functions. The primary reconstruction rates using DNNDAT with different loss functions are shown in Figure 33. Using the DNNDAT method in the same work area or using

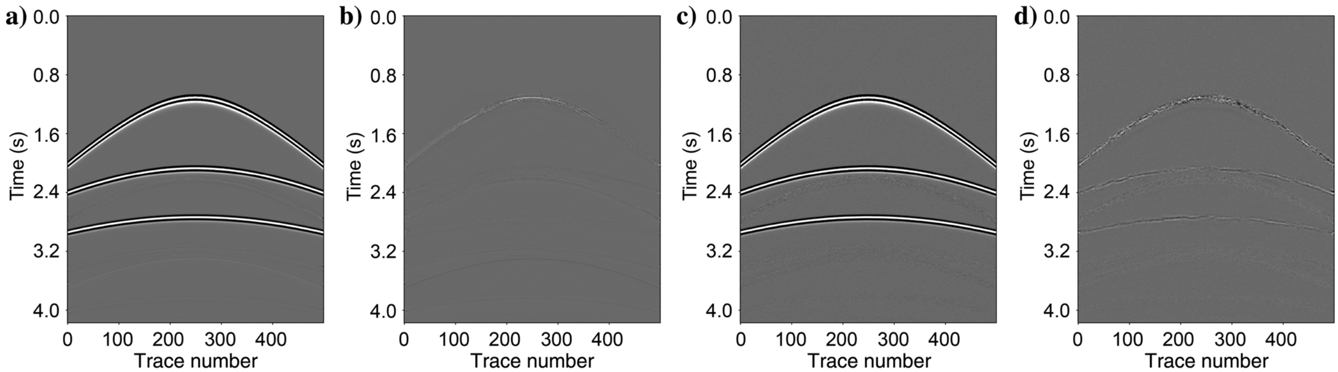

Figure 27. Reconstructed results and the corresponding differences using DNN trained by data without rotation. (a) Primaries reconstructed from clean full wavefield. (b) Difference between true primaries and (a). (c) Primaries reconstructed from noisy full wavefield. (d) Difference between true primaries and (c).

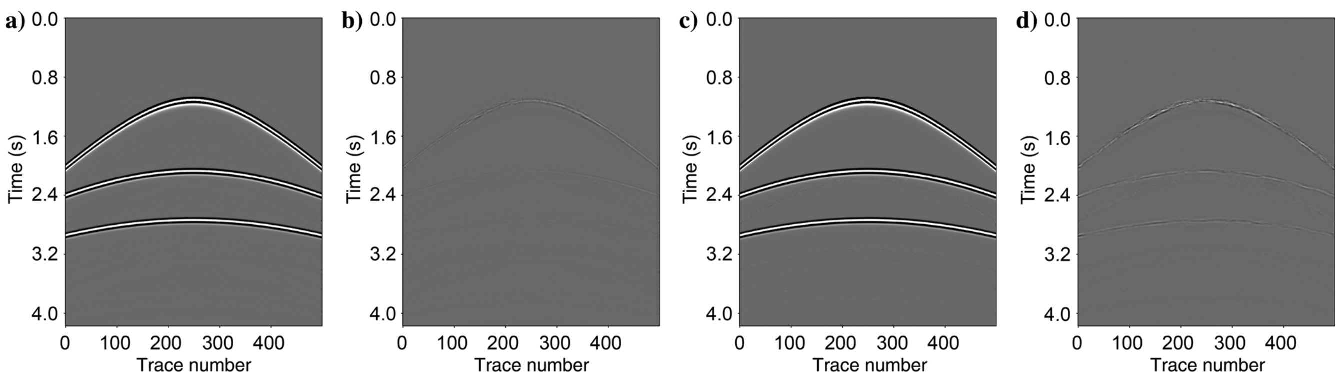

Figure 28. Reconstructed results and the corresponding differences using DNN trained by data without the predicted multiples. (a) Primaries reconstructed from clean full wavefield. (b) Difference between true primaries and (a). (c) Primaries reconstructed from noisy full wavefield. (d) Difference between true primaries and (c).

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DNNDATTL across the work areas to suppress multiples in the clean and noisy full wavefields, the primary reconstruction rates obtained with the MAE loss function are the largest, followed by the SSIM loss function. Multiple suppression with the Huber loss functions is closely related to the selection of the hyperparameter. Under the hyperparameter selected in Figure 33, the suppression of multiples using Huber functions is not as effective as those of the MAE function. This demonstrates that a more robust MAE can enable the DNNDAT to obtain a better primary reconstruction effect when multiples and background noise are outliers.

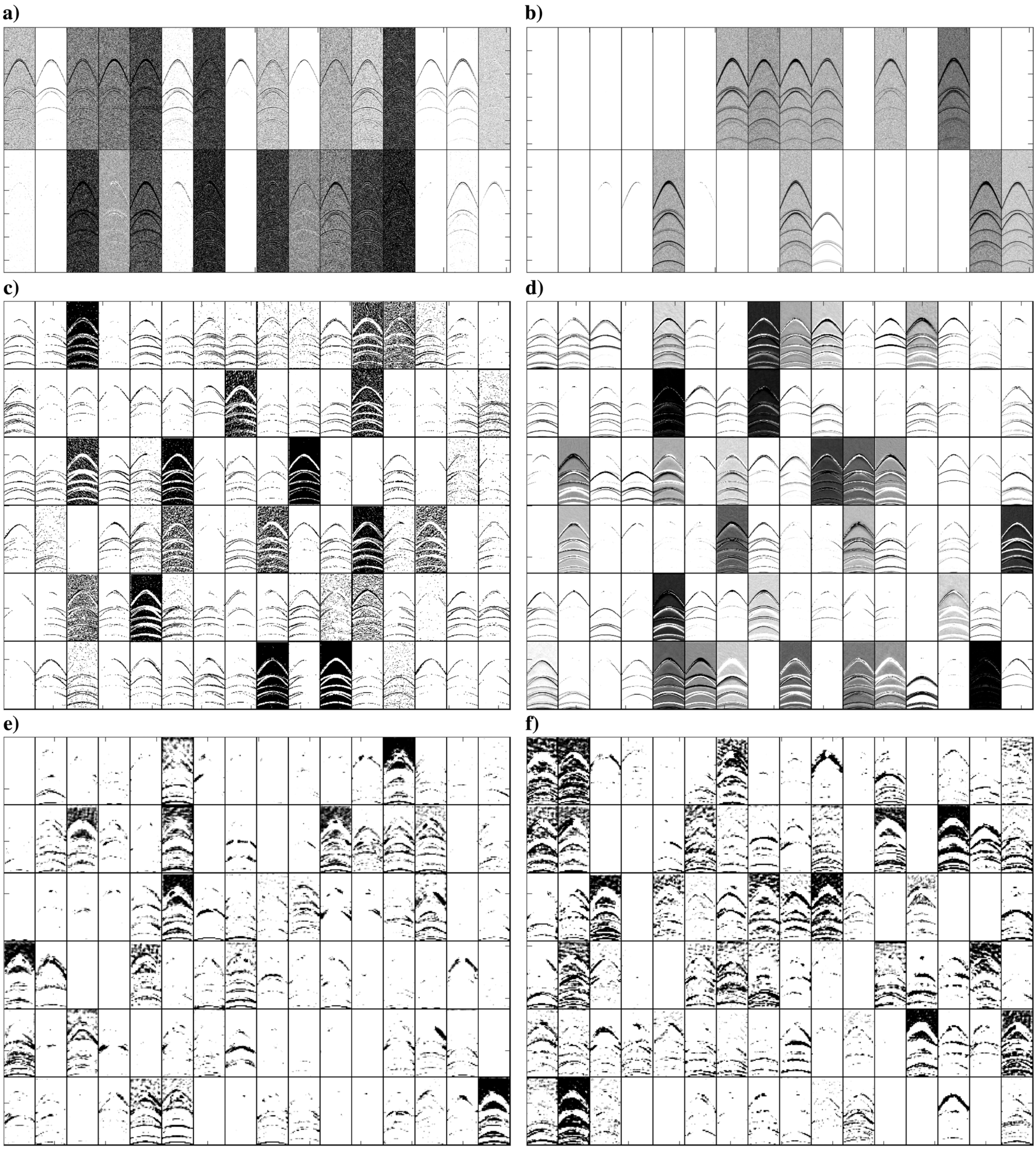

Figure 29. Visualization results. (a-f) Partial channels activated by the V1–V6 layers in the DNN trained without the predicted multiples, respectively.

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Robust reconstruction of primaries

When using the proposed DNNDAT method to suppress multiples and reconstruct primaries, background noise with different intensities is added to the training data set. We want to know up to what S/N can the DNNDAT make robust reconstruction for primaries. Taking the first simple data as an example, we add background noise with an S/N higher than −6 dB to the input data of the training data set and train the DNN to obtain the DNN parameters. Background noise with an S/N distribution of −14 to 2 dB is added to the test data and sent into the trained DNN to obtain the primary reconstruction rates, as shown in Figure 34. When the S/N of the test data is higher than −9 dB, the primary reconstruction rate is higher than 0.8, and the DNNDAT shows good robustness. The robustness of the DNNDAT is related to the noise intensity distribution added during the DNN training stage.

Future work

Peg-leg and internal multiples are more complex than free-surface-related multiples. However, the DNN framework and training method in this study also can be used to eliminate peg-leg and internal multiples. It should be noted that unlike the proposed suppression method for the free-surface-related multiples when removing peg-leg and internal multiples, the SRME method cannot be used to determine the predicted multiples that are used for the second channel of the input data. The model-based water-layer demultiple (Wang et al., 2014) and virtual events (Ikelle, 2006; Liu et al., 2018a) methods can be used to obtain the predicted multiples and preprocessed label data required to suppress the peg-leg and internal multiples, respectively.

The proposed method requires estimation of the predicted multiples using traditional multiple suppression methods (e.g., the SRME method) as one of the input data, which is time-consuming. However, the predicted multiples can improve the ability of the DNN to identify and suppress free-surface-related multiples. In addition to the predicted multiples, other data that can calibrate the true multiples also can train the DNN and act as a channel for the input data. When we have data from a variety of velocity models, a feasible solution to reduce computational overheads may be to combine the velocity models to train the DNN; that is, the velocity models may act as a channel for the input data.

The DAT method used in this study includes rotating the training data and adding background noise to the input data. In the field of image processing, random occlusion of images (Zhong et al., 2020), linear interpolation (Zhang et al., 2017b), and pixel fusion of two images at a certain ratio (Hendrycks et al., 2019) are all good data augmentation methods. To further increase the amount of data in the training data set, we can add swell noise to the marine training data set and surface waves or linear noise to the land training data set, as DAT methods. In addition, increasing the amount of training data by combining multiple velocity models may have positive results.

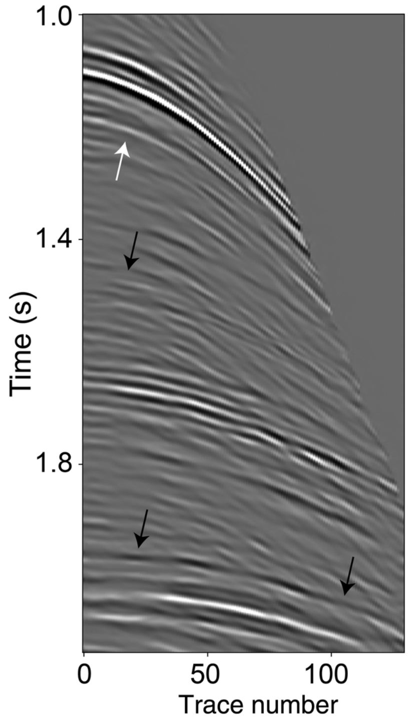

Figure 31. Reconstructed primaries using the DNN trained from scratch using the first three synthetic data sets. The white arrow denotes substantial residual multiples, and the black arrows identify substantial leakage and damage to primaries.

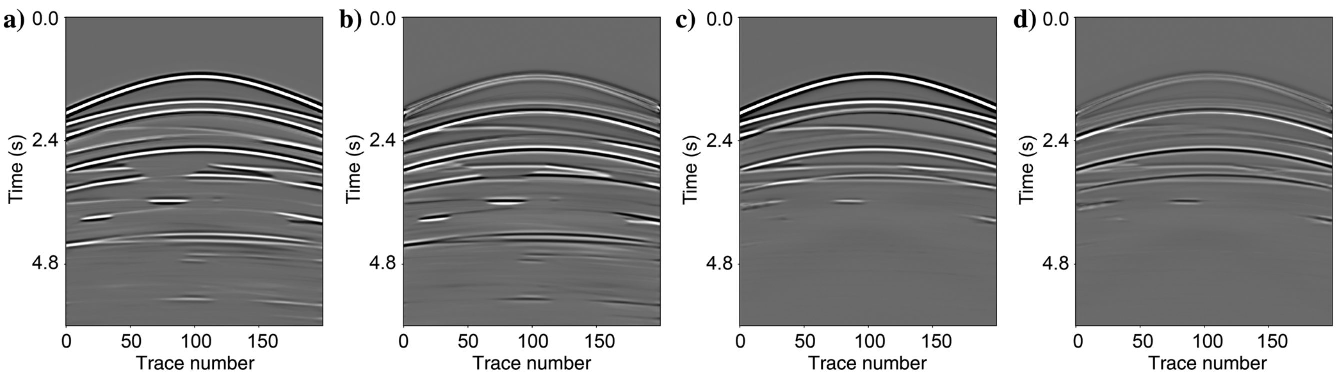

Figure 30. Reconstructed results and the corresponding differences for the second simple model data using the DNN trained by the first simple model data. Note that transfer learning is not used. (a) Reconstructed primaries using the DNN without the channel of predicted multiples. (b) Difference between true primaries and (a). (c) Reconstructed primaries using the DNN with the channel of predicted multiples. (d) Difference between true primaries and (c).

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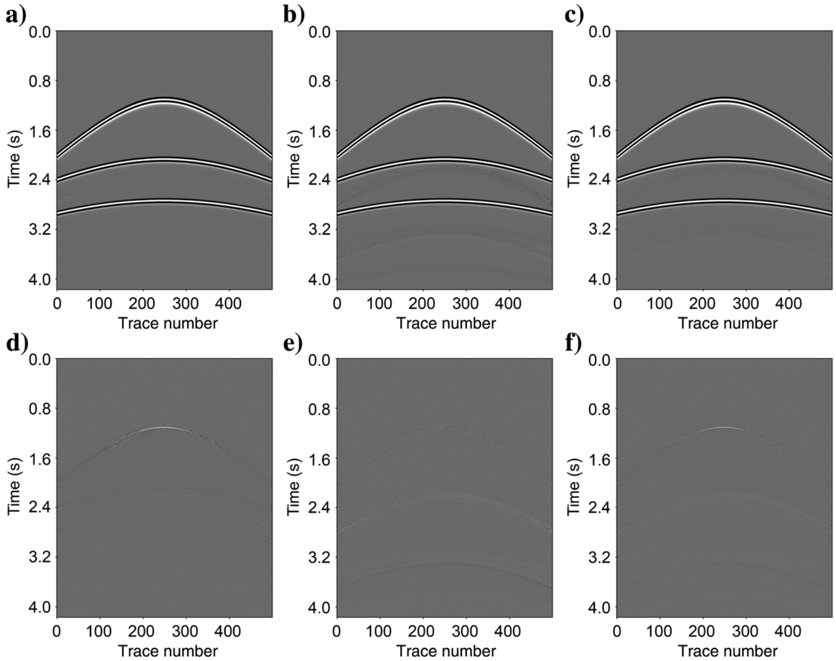

Figure 32. (a-c) Reconstructed primary results from the clean full wavefield of the first simple model using the DNN trained by training data without adding background noise and combined with the **P**-DAT1, **P**-DAT2, and **P**-DAT3 modes, respectively. (d-f) The corresponding differences between the true primaries and (a-c), respectively.

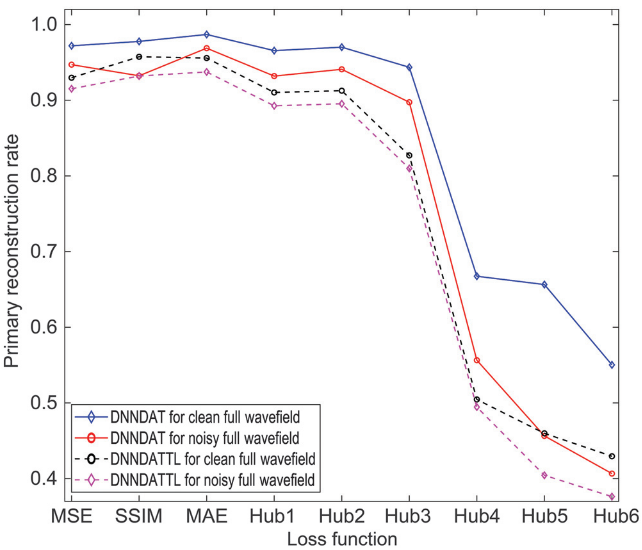

Figure 33. Primary reconstruction rates using the DNNDAT and DNNDATTL methods for clean and noisy full-wavefield data under different loss functions. The Hub1–Hub6 represent Huber loss functions with hyperparameters 1, 0.1, 0.01, 0.001, 0.0001, and 0.00001, respectively.

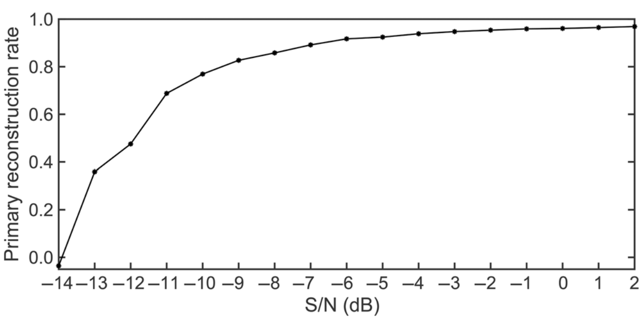

Figure 34. Primary reconstruction rates using test data with different S/Ns.

Obtaining a better multiple suppression effect using the DNNDAT based on smaller sample data sets requires further research.

CONCLUSION

A DNNDAT is used to reconstruct primaries and suppress multiples. The DNN structure designed is similar to that of the U-net, including the convolutional encoding and decoding parts. Considering that the field data contain a lot of background noise, we train the DNN using augmented training data sets to improve the robustness and generalization ability of the DNN, increase the size of the training data set, and reduce the risk of overfitting. This is performed by rotating and adding background noise of different intensities into the input data. During the testing stage and using the trained DNN parameters, convolutional encoding extracts features from the primaries, multiples, and background noise in the seismic data, whereas convolutional decoding uses the extracted features to reconstruct the primaries, multiples, and background noise. The transfer learning method improves the generalization ability of the DNN and enables the DNN parameters trained in one work area to be used for suppressing multiples in other work areas, achieving an effect that is comparable to traditional methods that can be used to obtain label data. Three sets of synthetic data (two simple models and the Pluto model) examples verify the effectiveness, efficiency, and stability of the proposed method for primary reconstruction and multiple suppression. Subsequently, one field data example shows that the proposed method can be applied to suppress seismic multiples under complex conditions.

During multiple suppression, the training of the DNN occupies the main computational cost, and the trained DNN can be used to reconstruct the primaries quickly or in real-time. Compared with the DNNGT method, the DNNDAT method, which uses augmented training data set pairs to train DNNs, has better antinoise stability and multiple suppression effects. However, for the DNNDAT method, it is necessary to use the rotation technique during the data augmentation process to avoid the problem of difficult convergence.

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To improve the effectiveness of suppressing multiples, an MAE loss function with good robustness is a better choice for training DNNs.

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DATA AND MATERIALS AVAILABILITY

Data associated with this research are available and can be obtained by contacting the corresponding author.

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Kunxi Wang received a B.S. (2017) in exploration technology and engineering from the Taiyuan University of Technology, Taiyuan, China, and an M.S. (2020) in solid geophysics from the Innovation Academy for Precision Measurement Science and Technology, Chinese Academy of Sciences, Wuhan, China. He is currently pursuing a Ph.D. from Peking University, Beijing, China. His research interests include deep learning, multiple suppression, simultaneous-source deblending, and denoising of seismic data.

Tianyue Hu received a B.S. (1983) and an M.S. (1986) in geophysics from the University of Science and Technology of China, Hefei, China, and a Ph.D. (1995) in geophysics from the University of London, London, UK. He was with the University of Chinese Academy of Sciences, Beijing, China, from 1996 to 1999. Since 2002, he has been a professor of geophysics at Peking University, Beijing. His current research interests include the methods of multiple attenuation and image, statics near surface, seismic interferometry, seismic modeling, and well ties. He is a member of SEG and EAGE.

Shangxu Wang received a B.S. (1984) in geophysics from the Wuhan College of Geoscience, Wuhan, China, and an M.S. (1987) and a Ph.D. (1990) in geophysics from the China University of Petroleum, Beijing, China. He has been with the China University of Petroleum since 1990. His research interests include seismic wave propagation theory, seismic forward modeling and inversion, and seismic-involved rock physics. He is a Changjiang River Scholars distinguished professor, the deputy director of the State Key Laboratory of Petroleum Resource and Prospecting, and the director of the Geophysics Key Laboratory Academic Committee, China National Petroleum Corporation, Beijing, China.

A biography and photograph of the other author are not available.

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